

# Logistics Performance, Ratings, and its impact on Customer Purchasing Behavior and Sales in E-commerce Platforms

Vinayak Deshpande

Kenan-Flagler Business School, UNC Chapel Hill, Chapel Hill, NC 27599, vinayak\_deshpande@kenan-flagler.unc.edu

Pradeep Pendem

Charles H. Lundquist College of Business, University of Oregon, Eugene, OR 97403, pradeep@uoregon.edu

**Problem Definition.** We examine the impact of logistics performance metrics such as delivery time, and customer's requested delivery speed on logistics service ratings and third-party sellers' sales on an e-commerce platform. **Academic/Practical Relevance:** While e-commerce retailers like Amazon have recently invested heavily in their logistics networks to provide faster delivery to customers, there is scant academic literature that tests and quantifies the premise that convenient and fast delivery will drive sales. In this paper, we provide empirical evidence on whether this relationship holds in practice by analyzing a mechanism that connects delivery performance to sales through logistics ratings. Prior academic work on online ratings in e-commerce platforms has mostly analyzed customers' response to product functional performance and biases that exist within. Our study contributes to this stream of literature by examining customer experience from a service quality perspective by analyzing logistics service performance, logistics ratings, and its impact on customer purchase probability and sales. **Methodology.** Using an extensive data set of more than 15 million customer orders on the Tmall platform and Cainiao network (logistics arm of Alibaba), we employ the Heckman ordered regression model to explain the variation in customers' rating of logistics performance, as well as the likelihood of customers posting a logistics rating. Next, we develop a generic customer choice model that links the customer's likelihood of making a purchase to the logistics ratings provided by prior customers. We implement a two-step estimation of the choice model to quantify the impact of logistics ratings on customer purchase probability and third-party seller sales. **Results.** We surprisingly find that even customers with no promise on delivery speed are likely to post lower logistics ratings for delivery times longer than two days. Although these customers are not promised an explicit delivery deadline, they seem to have a mental threshold of two days and expect deliveries to be made within that time. Similarly, we find that priority customers (those with two-day and one-day promise speed) provide lower logistics ratings for delivery times longer than their anticipated delivery date. We estimate that reducing the delivery time of all three-day delivered orders on this platform (which makeup  $\approx 35\%$  of the total orders) to two days would improve the average daily third-party seller sales by 13.3% on this platform. The impact of delivery time performance on sales is more significant for sellers with a higher percentage of three-day delivered orders and a higher spend per order. **Managerial Implications.** Our study emphasizes that delivery performance and logistics ratings, which measure service quality, are essential drivers of the customer purchase decision on e-commerce platforms. Further, by quantifying the impact of delivery time performance on sales, our study also provides a framework for online retailers to assess if the increase in sales due to improved logistics performance can offset the increase in additional infrastructure costs required for faster deliveries. Our study's insights are relevant to third-party sellers and e-commerce platform managers who aim to improve long-term online customer traffic and sales.

*Key words:* E-commerce, Integrated Warehousing and Logistics, Logistics Ratings, Big Data Analytics

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## 1. Introduction

E-commerce is one of the largest growing sectors in the digital economy, with sales of \$766.77 billion in the United States in 2019 and compound growth of 20.8% during 2014-2019 (IBISWorld 2019). Online retailers provide several advantages for a customer over traveling to a brick-and-mortar retail store, such as a more comprehensive selection of products and the ability to find desired products quickly. With the explosion of digitization and the added convenience of access to mobile devices, the industry expects to reach sales of \$1,916 billion in 2024 with compounded growth of 20.1% from 2019-2024 (IBISWorld 2019).

With the belief that fast and convenient delivery will attract more new customers, online retailers started to offer premium membership subscription, which typically promises fast shipping. For example, Amazon launched a premium service in 2005, Prime, with a promise of two-day shipping as the main draw. Fast forward to 2020; the company now has more than 150 million customers around the world who subscribe to Prime membership service (Del Rey 2020). Amazon has continued innovating and investing significantly in fast shipping advancements to improve customer service experience. According to Jeff Bezos, the number of items delivered to US customers with Prime's free one-day and same-day delivery more than quadrupled during 2018-2019 (Del Rey 2020). A similar trend towards fast shipping services has spread across some of the US's top retailers, such as Walmart, Target, and Costco (Montasell 2020). For example, in response to competition from Amazon, Walmart launched a two-day shipping service in June 2016, and three years later, it started offering next-day shipping in May 2019 (Crook 2016, Perez 2019). Similarly, Target launched free two-day and same-day deliveries in October 2018 and June 2019, respectively (Liptak 2019, Perez 2018). Overall, market trends show that big online retailers invest heavily to improve the customer service experience by reducing delivery time.

Although faster shipping service is convenient and provides superior service experience to customers, it has significant cost implications for online retailers. The additional capacity investments required to extend the reach of faster delivery service range from expanding fulfillment centers, distribution centers, air-cargo operations to the rental fleet. For example, Amazon spent \$21.7 billion on shipping costs in 2017, nearly twice what it spent in 2015 (Semuels 2018). Amazon's worldwide shipping costs jumped 46% to \$9.6 billion in the third quarter of 2019 from the previous year related to its one-day shipping program for Prime subscribers (Semuels 2018). The company profits slipped by 26% in October 2019 due to high shipping costs on one-day deliveries for Prime customers (Mattioli 2019).

With increasing e-commerce demand and advancements in fast shipping, online retailers are likely to incur higher shipping costs in the future. As a result, it is of primary interest to online retailers such as Amazon to understand if the increasing costs incurred in fast shipping provide a significant return

in additional sales. Although there is anecdotal evidence that online sales growth has accelerated with more investments in one-day shipping (Herrera and Qian 2019), there is scant academic literature and no formal evidence that faster shipping has given an additional boost to sales. In this paper, we focus on quantifying the value of fast shipping by examining the impact of delivery time reduction on the online platform's sales. In particular, we provide empirical evidence on whether this relationship holds in practice by analyzing a mechanism that connects delivery performance to sales through logistics ratings.

From a customer's perspective, products on most online platforms are sold by two kinds of sellers: (i) The platform itself as the seller. For example, Amazon sells products on its platform, (ii) Independent third-party sellers who sell on the platform. We channel our effort to particularly examine the third-party sellers as they have proliferated in recent years. The proportion and share of revenue of third-party sellers on Amazon have grown from 25% to 52% during 2007-2018 (Richter 2019, Clement 2020). This leads to our main research question: What is the impact of fast shipping or reducing delivery time on sales for a third-party seller on the e-commerce platform? (**Q1**). We identify the mechanism behind the relationship between delivery performance and sales in two phases. In the first phase, we establish the link between delivery performance and online logistics ratings, and in the second phase, we link logistics ratings to online sales, as described below.

In the current digital age, sellers can now offer a wide variety of products and different versions (e.g., model, style, color) of a single product. The rich information available about products means that customers can choose from a wide range of options. To ease the search cost, reduce product uncertainty (Chen and Xie 2008), and help customers in the purchase decision process, most platforms provide ratings of products and sellers in addition to information such as price, discounts, and available inventory. The ratings are typically displayed in the form of a distribution ranging from 1 to 5, with 1 being the lowest quality and 5 being the highest quality. Sellers rely heavily on ratings to maintain market share and survive against fierce price competition. Platforms typically display ratings on two dimensions on quality viz. product and logistics service. Product quality signifies the response to customer's experiences on the product performance, as stated on their web page. On the other hand, the logistics service quality typically signifies the response to the customer experience of timely order delivery. A customer with an intent to purchase visits the platform obtains price, ratings, and other sources of information on the product page. The customer then evaluates price, ratings across multiple sellers and finally decides on product purchase. At the end of the purchase process, sellers generally provide different shipping speed options: (i) fast delivery, or (ii) regular delivery. At this point, the customer decides on the choice of shipping speed and places the order. The selection of shipping speed by the customer sets an expectation of logistics service quality for the seller. The customer later receives the product and evaluates both the actual product and logistics

service quality. If the product functionality conforms to the web page's specification, the customer likely provides a high product quality rating (Chintagunta et al. 2010, Lin et al. 2011). On the other hand, when the seller delivers the product either on-time or earlier, the customer is likely to be elated with the logistics service experience. However, it is unclear if their response after a satisfying service experience manifests in a higher logistics rating. This leads to the first phase of our main research question: What is the impact of delivery time on the logistics rating provided by the customer? (Q1a).

Higher ratings about the seller and their products are likely to increase the likelihood of an incoming customer's purchase. So, it is no surprise that e-commerce platforms make enormous efforts by sending emails or text messages requesting feedback about their products and service from previous customers. A cumulative volume of higher logistics ratings from customers over time increases the effective rating visible to an incoming customer. A higher overall logistics rating is then likely to increase the likelihood of a purchase, which affects the seller's long-term traffic and sales. Prior work on online product quality ratings has shown evidence of their impact on customer purchase probability or sales (Chevalier and Mayzlin 2006, Chintagunta et al. 2010, Lin et al. 2011). However, the answer to whether a higher logistics rating impacts customer purchase decisions is unclear. This forms the basis for the second phase of our main research question: What is the impact of improved logistics ratings on customer purchase probability and sales for the seller? (Q1b)<sup>1</sup>.

In summary, we analyze the mechanism by examining the following two research questions individually. First, what is the impact of reducing delivery time on the logistics rating provided by a customer? (Q1a). Second, what is the impact of improved logistics ratings on customer purchase probability and sales for the seller? (Q1b). We then combine our results from Q1a and Q1b to answer our main research question: What is the impact of reducing delivery time on sales for a third-party seller on the e-commerce platform? (Q1).

The rest of the paper is organized as follows. In §2, we provide details on prior work on logistics performance, online ratings and highlight our contribution to the literature. In §3, we briefly describe the study setting, Cainiao's logistics operations, and summarize the data. In §4, we provide the measures, model, and results on the impact of reducing delivery time on the logistics rating of a seller. In §5, we build a generic choice model to examine the impact of improved logistics ratings on customer purchase probability and provide estimation results. In §6, we utilize the results derived from §4 and §5 to quantify the impact of reducing delivery time on seller sales. In §7, we examine

<sup>1</sup> In examining the impact of logistics ratings on customer purchase probability, we primarily consider focal customer's responses to feedback from accumulated ratings by other customers or Word-Of-Mouth (WOM) (Dichter 1966, Hennig-Thurau et al. 2004). We exclude the customer's responses based on their recent and past service experience due to a limited number of such transactions in our data. We gathered the count of orders for each unique customer and seller combination in our data. We found the median and 95th percentile of the count to be 1. These statistics indicate that most Cainiao customers have a single transaction with any seller in our data, which points to limited prior experience for any customer-seller pair.

the robustness of the causal link between delivery performance, customer-provided logistics rating, and third party seller's sales by addressing one form of independent variable endogeneity. Lastly, in §8, we conclude our work with managerial insights and highlight the limitations of our study.

## 2. Literature

Our work contributes to the following four areas of research: (i) Relationship between logistics performance and financial outcomes, (ii) Drivers of ratings on online platforms, (iii) Effect of disconfirmation on customer satisfaction, (iv) Customer choice models under censored choices and limited (or no) market size information. We discuss the prior literature in each of these areas and identify the contribution of our paper.

**Relationship between logistics performance and financial outcomes.** Recent literature in OM has looked at empirical evidence on the relationship between logistics performance and financial outcomes. Allon et al. (2011) estimate that a seven-second reduction in customer's wait time at a fast-food drive-thru chain can result in an average 1-3% increase in market share. Cui et al. (2019) finds that removing a high-quality logistics carrier option for a large online retailer leads to a decrease in sales by 16.42%, while its resumption increases by 18.83%. Fisher et al. (2019) extrapolate how faster delivery affects sales in the online channel of a US apparel retailer by leveraging a quasi-experiment involving the opening of a new distribution center, which resulted in unannounced faster deliveries to the western US states through its online channel. Unlike Fisher et al. (2019), who used aggregated average estimates of delivery time, we use actual observed delivery times to individual customers to identify their response to delivery time performance. The mechanism of improved financial outcomes from superior logistics performance can be a result of a customer's response to: (i) their own recent or past service experience, or (ii) to accumulated feedback from previous customers' experience on the common platform (Word-of-Mouth). All the studies mentioned above attribute the improvement to the first mechanism (customer's own prior experience), but do not empirically validate this mechanism. In a recent study, Mao et al. (2019) find that a ten-minute earlier delivery from the targeted (or expected) delivery time increases each customer's future demand by 1.03 orders per month on an on-demand meal delivery service platform. Like the above studies, this work attributes the increase in sales from superior delivery performance to the first mechanism and, moreover, provides an empirical validation of this mechanism.

Unlike the prior studies, our research focuses on the second mechanism (word-of-mouth effect through logistics ratings) by providing an empirical validation of this mechanism. In summary, our study contributes to this stream of literature on quantifying the impact of faster delivery on financial outcomes by analyzing actual delivery times on individual orders to identify the mechanism of customer response to accumulated feedback on delivery performance by other customers.

**Drivers of ratings on online platforms.** The work on online ratings in the Marketing or Information Systems literature has mostly focused on physical products (e.g., Books, DVDs, and videos). The ratings for these products on platforms signify product functional quality. Most of the prior work on ratings has primarily focused on different biases influencing the ratings/score. These biases can originate in various forms ranging from the level of an individual customer to the firm. Some of the biases that arise at a personal level include self-presentational behavior (Schlosser 2005), self-selection (Li and Hitt 2008, Hu et al. 2009), previous opinions (Wu and Huberman 2008), high-quality sellers (Li and Xiao 2010). Social biases comprise identity (Wang 2010), comparisons (Chen et al. 2010), high valance, and high volume rating environments (Moe and Schweidel 2012), friends and strangers (Lee et al. 2015), cultural influences (Koh et al. 2010). At the organizational level, Dellarocas (2006) showed that ratings might not be wholly trusted as they could be strategically manipulated by firms (potentially inflated) to remain competitive or maintain market share.

Our work differs from previous research as we study logistics service quality rather than product quality. Specifically, the ratings that we consider are the customer's response to an experience of logistics service-fulfillment from the point of purchase to delivery rather than product performance quality. The closest work on logistics ratings in OM is by Bray (2020), where the author shows that scores are higher when the track-package activities cluster toward the end of the shipping horizon than at the start. Our contribution to this stream of literature is primarily on examining the impact of observable (both to the seller and customer) operational factors on logistics ratings, which has not been studied previously.

**Effect of disconfirmation on customer satisfaction.** The Expectation-Disconfirmation theory (Oliver 1977, 1980) in service marketing is a prominent theory applied to understand drivers of customer satisfaction. The theory states that disconfirmation of product or service quality affects customer satisfaction. Disconfirmation is the discrepancy between customer expectation and the actual performance of product or service quality. Each customer builds a certain level of expectation for the product or quality before initiating the purchase. After receiving the product, the customer consumes it and experiences the performance of its quality. If the performance is worse than expected, the customer experiences more discomfort, which leads to more dissatisfaction, thereby resulting in lower ratings. In online platforms, the customer's actual expectation of quality is not explicitly specified by the customer and is unknown to the seller. Similarly, the performance after consuming the product is only observed by the customer and is unknown to the seller. The response to performance compared with their expectation manifests in the form of a customer's rating. As a result, research that examined the impact of disconfirmation on customer satisfaction has considered the distribution of ratings before the purchase, and the rating posted by a customer as a proxy for expectation and the disconfirmation (Anderson and Sullivan 1993, Bhattacharjee 2001, McKinney et al. 2002, Susarla

et al. 2006, Ho et al. 2017b). A different stream of literature on online ratings also notes that the distribution is often driven by different biases, such as purchasing bias and under-reporting bias (Hu et al. 2009). As a result, the measure of discomfort based on the prior distribution of ratings and its impact on customer satisfaction is likely biased.

Like Bray (2020), our study contributes to this stream of literature by proposing an accurate measure of discomfort and its impact on customer satisfaction. In our study, a customer explicitly states their expectations (choice of shipping speed) and performance (delivery time) is observed both by the customer and seller. Hence, the disconfirmation derived from the selection of shipping speed and delivery time generates an accurate measure of discomfort. Further, our study also contributes to understanding the effects of discomfort on a customer's likelihood of posting a rating by different customer types. Ho et al. (2017b) finds that an individual is more likely to post a review when the magnitude of disconfirmation (s)he encounters is larger. Our study also finds a similar result for customers with no promise on delivery speed and, hence, extends our understanding of the effect of discomfort for customers.

**Customer choice models under censored choices and limited (or no) market size information.** Customer choice models describe the decision maker's choices among alternatives derived under the assumption of utility-maximizing behavior (Train 2009). The alternatives (referred to as the choice set) represent competing products or sellers over which choices are made. The modeling framework allows the choice decision to be related to explanatory variables. The simplicity of these models allows for the derivation of choice probability expressions without reference to how the choice is made. The model parameters are then estimated by equating the choice probabilities to observed market shares based on sales information. The simpleness of the framework and the expressions comes with critical information requirements without which estimation can result in biased and inconsistent parameters. One such information is the availability of an exhaustive list of competitors, sales, price, and market size information. Often due to data limitations, only a subset of such information is available. For example, when an exhaustive list of competitors is unavailable or partly available, the general approach is to put those competitors into an outside option (Vulcano et al. 2012, Newman et al. 2014) or use competitor stockouts and price variations (Fisher et al. 2018). Similarly, when market size (number of potential customers) information is unavailable, the approach is to develop expressions free of market size by integrating across all possible values (Vulcano et al. 2012, Newman et al. 2014) or assuming a market size based on demographic information (Berry et al. 1995).

In our context, we face the same challenges where we have data on products, their attributes, and sales information only from sellers operating on a single platform. It is likely possible that an incoming customer can access similar products from other channels such as competing online platforms (e.g., JD.com) and brick-and-mortar stores. In the standard MNL model (Train 2009),

the customer compares the utility received from different alternatives and decides to choose one of them or do not purchase at all (typically referred to as the “outside option”, and its utility is assumed to be 0). Because the sellers on the platform that we study do not form an exhaustive list of competitors, customers can receive non-zero utility of purchase from other channels, thus violating the assumption of zero utility for the outside option. We utilize the framework in Newman et al. (2014), which proposes an efficient and consistent estimation of choice-based models when one alternative is completely censored. Other channels of purchase, such as competing online platforms and brick-and-mortar stores, are entirely censored in our context. We build on this choice model and extend it to incorporate information on actual market size through the number of unique customer visits to the platform. Further, we allow utility specification of the “no-purchase” alternative to vary by time, which has not been considered in the prior literature.

### 3. Study Setting & Data

We utilize data from the Cainiao network and the Tmall platform provided by the 2018 MSOM data-driven research challenge to study the research questions stated in the paper. A brief description of the study setting is as follows.

The Cainiao Network is a consortium of warehouse and carrier companies founded by Alibaba. The network operates as integrated warehousing and logistics platform linking storage, distribution centers, and carrier providers. Tmall is a B2C Chinese language online retail website operated by Alibaba since April 2008. As of February 2018, the site had over 500 million monthly active users. Before forming the network in 2013, sellers on Tmall managed the inventory and fulfilled orders by themselves or by using various third-party carriers. The Cainiao network opened its service to all the sellers on their platform who now have the option to have their inventory managed and customer orders fulfilled conveniently.

The life cycle of a customer order on the Tmall platform from its inception to delivery is as follows. A customer visits the platform on a personal computer or a mobile app. Next, (s)he browses the products from different sellers and places the order for a seller on the platform (“order arrival” event in our data). Sellers on the platform are classified into two types based on whether they depend on the Cainiao network for management of product inventory and fulfillment of order delivery. The first type of sellers are independent of Cainiao and manage the inventory of products in their warehouse. These sellers are responsible for shipping the product from their warehouse to the carrier location. The second type of sellers are dependent on Cainiao for managing the product inventory. Cainiao is responsible for shipping these products to the carrier location. The package is next shipped (“consignment” event in our data) to the carrier location. The carrier then sends the package via multiple logistics transfers (e.g., air, rail, and road) to the delivery station near the customer location.

The package is then finally shipped from the delivery station to the customer's location by the carrier. The customer acknowledges receipt of the package by providing a signature ("signed" event in our data) to the delivery person. Later, the customer may decide to post a rating for their experience on the platform, including their experience on order quality, online purchase experience, and the logistics service quality.

We utilize information on customer orders, a detailed timeline of delivery events for each of these individual orders, and information about unique customer visits to the platform from April 2017 for our analysis. The customer order data possesses granular information including the date and time-stamp of order arrival, customer identifier, item identifier, quantity ordered, total payment in Chinese yuan, customer's requested delivery speed (can take three unique values: no-promise speed, two-day promise speed, and one-day promise speed), Cainiao indicator (1: if the order used Cainiao warehousing and logistics service, 0: otherwise), seller identifier, carrier identifier, and logistics rating (customer's response to the logistics service, uses a Likert scale from 1 to 5 with 1 being low quality and 5 being high quality).

The logistics data describes the event timeline from warehouse shipment to final delivery for each customer order. The time duration from the "order arrival" event to the "signed" event measures each customer order's delivery time. Although the data is highly granular, we do not have information on product categories (e.g., book or electronics), customer demographics (e.g., location of the customer, gender, or age), seller or carrier region, or marketplaces in the dataset provided by the MSOM data driven research challenge. Hence, our research cannot address questions related to heterogeneity arising from product category, customer type, region or marketplaces. Table 1 provides an overview of the data-set.

**Table 1** Cainiao network data: April 2017

| Variable          | Value                                                   |
|-------------------|---------------------------------------------------------|
| Customer orders   | 15,848,263                                              |
| Unique customers  | 13,951,407                                              |
| Unique sellers    | 345                                                     |
| Unique items      | 139,354                                                 |
| Total revenue     | 1.878 Billion Chinese Yuan ( $\approx$ \$0.270 Billion) |
| Carrier companies | 14                                                      |

A total of 13,951,407 unique customers placed 15,848,263 orders during April 2017. We focused on the top 14 carrier companies, which handled 95% of the total volume for our analysis. The largest carrier company, identified as "184" in the data, delivered more than 25% of total customer orders. We found that each customer order had a median paid amount of 44.6 yuan.

Table 2 lists the distribution of the customer's requested promise speed, delivery time, and logistics rating. We find a significant number (91.8%) of orders had no promise speed, meaning these orders

have no time restriction on when the customer can expect to receive the package. The remaining 8.2% of the orders account for two-day and one-day promise speed requests. Around 95% of the orders are delivered within five days, and most orders are delivered in three days. We find that 62.2% of customer orders contained no logistics service rating in our data. The remaining 37.8% of orders with a logistics rating have a J-shaped distribution for the ratings, which is commonly observed on online platforms (Hu et al. 2009).

**Table 2** Distribution of Promise speed, Delivery time & Logistics rating

| Promise speed |            |            | Delivery time |           |            | Logistics rating |           |            |
|---------------|------------|------------|---------------|-----------|------------|------------------|-----------|------------|
| Status        | Count      | Proportion | Value (days)  | Count     | Proportion | Status           | Count     | Proportion |
| No-promise    | 14,546,267 | 91.8%      | 1             | 131,532   | 0.8%       | No response      | 9,849,239 | 62.2%      |
| Two-day       | 1,167,622  | 7.4%       | 2             | 2,819,844 | 17.8%      | 1                | 90,982    | 0.6%       |
| One-day       | 134,374    | 0.8%       | 3             | 5,597,409 | 35.3%      | 2                | 31,902    | 0.2%       |
|               |            |            | 4             | 4,548,837 | 28.7%      | 3                | 134,372   | 0.8%       |
|               |            |            | 5             | 1,825,257 | 11.5%      | 4                | 356,404   | 2.2%       |
|               |            |            | $\geq 6$      | 925,384   | 5.8%       | 5                | 5,385,364 | 34.0%      |

## 4. Impact of Delivery Time on Logistics Rating

In this section, we analyze the first phase of our research question: What is the impact of delivery time on the logistics rating provided by the customer for the seller from whom they purchased the product? (Q1a).

### 4.1. Measures

Using each customer order as the unit of analysis, we aim to understand the impact of different operational variables such as delivery time, total order payment, promise speed of delivery, and additional variables on the logistics rating. The detailed definitions for each of these measures, which go into the econometric model, are provided below.

**Logistics rating.** Logistics rating (the dependent variable) is the customer's rating for their experience with the logistics service for their order. This variable uses a Likert scale with values ranging from 1 (low quality) to 5 (high quality). We find that 62.2% of the customer orders have a "no response" for the rating in Table 2. Excluding these observations, which are substantial in number, can potentially lead to biased estimates. Hence, we provide an econometric specification which models "no response" option, as described later.

The independent variables are as follows.

**Delivery time.** Delivery time is the time duration from the moment a customer placed the order to the time s(he) received the package. We measure the unit of this variable in "days", which is the same unit of measure as the customer's requested promise speed.

**Pay norm.** Pay norm is the scaled measure of the total amount paid for each customer order. This variable is scaled to a mean 0 and standard deviation 1. We use this scaled measure rather than the actual value for two reasons: (i) the variation in the total amount paid in the data is relatively large (minimum 0.01 yuan and maximum 1,215,146 yuan), (ii) to prevent having a small number for the coefficient in the regression results.

**Cainiao.** Cainiao is a binary variable that takes the value of “1” if Cainiao managed the customer order, and the value “0” if managed by the independent seller.

**Controls.** We use several controls in our model since the underlying customer’s utility to post a rating (or no rating), and the ratings they provide from logistics service is likely to depend on additional variables. Our controls include time-invariant variables such as the seller from whom the product was purchased, the carrier company that delivered the order, and time effects such as the week of the month, day of the week, hour of the day, and holidays. Each seller and the carrier company may follow different inventory management and shipment policies, which are not directly observed in the data. Hence, we control for these omitted variables through the seller, carrier company fixed effects, and time effects.

## 4.2. Model & Results

We explain the underlying data generating process of observed logistics ratings. The dependent variable, logistics rating, is a discrete ordered outcome ranging from 1 to 5 with 1 being a low rating and 5 being a high rating. In our data set, we find 62.2% of the customer orders with rating status “no response”. Hence, our data sample is likely to suffer from the selection on unobservables problem (De Luca and Perotti 2011) because the errors that determine whether a customer posts (or does not post) a rating are potentially correlated with errors that determine the value of a rating. Next, we describe a model that addresses this issue.

There are two dependent variables in our data generating process. First, there is a binary outcome,  $s_{it}$  that indicates whether customer  $i$  in the sample has provided a rating. Second, there is an ordinal outcome, which is the actual rating provided by customer  $i$ ,  $Y_{it}$ , conditional on customer  $i$  posting a rating. Both outcomes are categorical. We model both outcomes jointly as the errors in their data generating process are likely to be correlated due to sample-selection. The outcomes are modeled as a linear combination of covariates relative to specific cutoff points that partition the real line. With idiosyncratic errors that are normally distributed, the econometric specification is a combination of the Probit selection model and the Ordered Probit rating model, referred to as the Heckman Ordered Probit Regression model (De Luca and Perotti 2011). An important feature of this model is that it does not discard no-rating response observations and models both the customer’s rating decision and the ratings that they provide in the data generating process. While the Bivariate probit model has

been used in prior operations literature (Kim et al. 2015), we are not aware of the application of the Heckman ordered probit model in an operations context. The model specification is as follows.

The selection model is given by

$$s_{it} = \begin{cases} 1 & \text{if } (\mathbf{X}_{it} \cdot \boldsymbol{\gamma} + \epsilon_{1it} > 0) \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The ordinal outcome model for the logistics rating,  $Y_{it}$ , is as follows. The probability that the ordinal outcome  $Y_{it}$  is equal to  $r$  is the probability that  $\mathbf{X}_{it} \cdot \boldsymbol{\beta} + \epsilon_{2it}$  falls between  $K_{r-1}$  and  $K_r$

$$\Pr(Y_{it} = r) = \Pr(K_{r-1} < \mathbf{X}_{it} \cdot \boldsymbol{\beta} + \epsilon_{2it} \leq K_r) \quad (2)$$

$Y_{it}$  is the ordinal outcome for order  $i$  placed at time  $t$ , while the vector  $\mathbf{X}_{it}$  comprises all covariates in the model.  $\boldsymbol{\gamma}, \boldsymbol{\beta}$  represent the coefficients of the selection model and ordinal outcome model, respectively. The variable  $r$  takes values 1, 2, 3, 4 & 5.  $K_0$  to  $K_5$  are cut-off points on the underlying customer latent response curve with  $K_0 = -\infty$  and  $K_5 = \infty$ .  $\epsilon_{1it}, \epsilon_{2it}$  are the idiosyncratic random error terms in respective models and are assumed to follow a bivariate normal distribution with mean 0 and correlation coefficient  $\rho$ .

The Heckman ordered model stated above, although new to the operations management field, is a commonly used model in marketing and management literature utilized to understand the drivers of customer-provided ratings. The selection model is often referred to as an Incidence model (Moe and Schweidel 2012, Lee et al. 2015, Karaman 2020) or Propensity model (Ho et al. 2017b). The ordered outcome model is referred to as Evaluation model (Moe and Schweidel 2012, Ho et al. 2017b) or Rating response model (Lee et al. 2015). A distinctive feature of the Heckman models is that they account for and help resolve the **sample-induced endogeneity** resulting from non-random sample selection such as truncated and censored samples (Certo et al. 2016). However, this model's primary limitation is that it does not address the endogeneity<sup>2</sup> resulting from independent variables (Certo et al. 2016).

We ran the Heckman ordered probit regression model on each subset of promise speed orders (no-promise, two-day promise, and one-day promise) separately. Table 3 provides the regression results of all the models. Column (1) lists all the covariates, controls, and model statistics. Across all the models, the reference level of the delivery time variable is "1 day". Columns (2) and (3) list the probit selection and ordered probit regression results on "No-promise speed" customer order data. From column (2), we find that the coefficient of delivery time increases with longer delivery time. This

<sup>2</sup> We analyzed and reported the results on multiple forms of endogeneity emanating from independent variables such as selection bias (selection on observables and unobservables), simultaneity, and omitted variable bias (Ho et al. 2017a) in §7.1 of the paper and §A of the online appendix. This analysis demonstrates and strengthens the evidence of the causal relationship between delivery performance and customer-provided logistics ratings.

result suggests that the likelihood of a no-promise speed customer posting a rating increases with an increase in delivery time. From column (3) (ordinal outcome model), we find that the coefficient of delivery time decreases with longer delivery time. This result suggests that ratings provided by no-promise speed customers decrease with an increase in delivery time. A no-promise speed customer does not have a set deadline for order delivery. The seller or carrier provider is allowed to deliver the package for these orders at their convenience. Surprisingly, we find that the chance that no-promise speed customers (non-priority customers) post a logistics rating is higher, and the value of logistics rating is lower with delivery time greater than two days.

Expectation-Disconfirmation theory (Oliver 1977, 1980) has been extensively used to study customer satisfaction, as discussed in the literature review section. In both columns (2) and (3), we find the coefficient of “2 days” delivery time to be insignificant. So, if we consider “2 days” delivery time as a no-promise speed customer’s hypothetical baseline expectation of logistics service, orders delivered in three days, four days, five days, and later denote increasing order of discomfort. As a result, our results can be inferred as follows. A no-promise speed customer is more likely to post a rating and provide a lower rating with increasing discomfort. Analyzing data on complete purchasing and rating activities on an e-commerce website, Ho et al. (2017b) finds that an individual is more likely to post a rating when the magnitude of disconfirmation (s)he encounters is large. The measure for disconfirmation in Ho et al. (2017b) is derived from the distribution of ratings before purchase. We find a similar result in the context of no-promise speed customers, but using an accurate measure for discomfort. Hence, our results strengthen the underlying theory and provide unbiased results on the impact of discomfort on customer satisfaction. Drawing on parallels from Expectation–Disconfirmation theory and Ho et al. (2017b), we surprisingly find that a no-promise speed customer has an average expectation of two-days delivery time, although they are not promised any specific delivery time.

There are two possible explanations for the two-day delivery time being a hypothetical baseline expectation of logistics service for no-promise speed customers. First, a two-day delivery has become a logistics service quality norm for most online retailers (Crook 2016, Liptak 2019, Del Rey 2020). As a result, customers will likely be dissatisfied with the service and provide a lower rating for delivery time beyond two days. Second, the alternate explanation is based on customers anchoring their delivery expectations at two days or a little longer than two days. The anchoring effect theory suggests that people make predictions about uncertain events by starting from an initial value and then continually adjusting to reach the final value (Tversky and Kahneman 1974, Epley and Gilovich 2002). In examining the impact of delivery performance on logistics rating, we primarily focus on customer’s response to their focal order delivery performance. Our study does not include or model the customer’s responses based on their recent and past service experience due to a limited number of such transactions in our data. Hence, an incoming no-promise speed customer is likely to anchor

the expectation of their focal order delivery performance based on the available shipping options of certainty on the platform. The Tmall/Cainiao platform provided only two fast shipping choices, one-day and two-day, during April 2017 (our data period). A no-promise speed customer neither pays an additional price for logistics service nor sets the deadline for order delivery. Hence, we believe these customers are most likely to anchor their delivery expectation at the slowest shipping option on the platform (two days) or a little longer (three days or four days, etc.). Although the possibility of heterogeneous anchoring expectations exists across customers, our results show that no-promise speed customers anchor their expectation at the slowest guaranteed shipping option of two days available on the platform.

The insights from both the “Two-day promise speed” and “One-day promise speed” models are as follows. From columns (5) and (8), we find delivery time has no impact on these customers’ likelihood of posting a rating. However, from columns (6) and (9), we find coefficients of delivery time decrease with longer delivery time beyond their anticipated delivery speed. This result suggests that ratings provided by these customers decrease with an increase in delivery time beyond their expected delivery date. We do not find strong statistical significance of the channel (Cainiao or seller) variable across the selection and ordered outcome models. This result indicates that the channel of delivery does not impact logistics ratings.

The Heckman ordered probit regression is a non-linear model whose coefficient estimates cannot be directly interpreted as marginal effects. From columns (2) and (3), (5) and (6) in Table 3, we find that the coefficient of “2 days” delivery time is statistically insignificant. As a result, we combine the reference level “1 day” and “2 days” delivery time category into a single category for these customers to generate marginal effects. Table 4 lists the average marginal effects of the delivery time for all the models. The estimates from column (2) are inferred as follows. If a three-day delivered order is delivered in one day or two days, it decreases the average probability of a no-promise customer posting a rating by 0.0094. For the same customers, the average probability of posting a rating of 1, 2, 3, or 4 decreases by 0.0026, 0.0008, 0.0032, and 0.0069, respectively, while the probability of posting a rating of 5 increases by 0.0135. The average marginal effects for “Two-day promise speed” and “One-day promise speed” data are interpreted along similar lines. These estimates show that reducing delivery time reduces the chance of a low rating while increases the chance of receiving the highest rating for logistics service.

Table 3 Heckman Ordered Probit Regression results

| Variable      | No-promise speed |                         |                       | Two-day promise speed |                         |                       | One-day promise speed |                         |                       |
|---------------|------------------|-------------------------|-----------------------|-----------------------|-------------------------|-----------------------|-----------------------|-------------------------|-----------------------|
|               | Variable<br>(1)  | Probit Selection<br>(2) | Ordered Probit<br>(3) | Variable<br>(4)       | Probit Selection<br>(5) | Ordered Probit<br>(6) | Variable<br>(7)       | Probit Selection<br>(8) | Ordered Probit<br>(9) |
| Delivery time | 2 days           | 0.034<br>(0.019)        | -0.048<br>(0.047)     | 2 days                | 0.055<br>(0.039)        | -0.170<br>(0.192)     | 2 days                | -0.051<br>(0.037)       | -0.350***<br>(0.035)  |
|               | 3 days           | 0.059**<br>(0.019)      | -0.134**<br>(0.046)   | 3 days                | 0.041<br>(0.039)        | -0.539**<br>(0.196)   | ≥ 3 days              | -0.041<br>(0.035)       | -0.350***<br>(0.105)  |
|               | 4 days           | 0.086***<br>(0.019)     | -0.210***<br>(0.046)  | ≥ 4 days              | -0.028<br>(0.039)       | -0.796***<br>(0.201)  |                       |                         |                       |
|               | 5 days           | 0.105***<br>(0.019)     | -0.295***<br>(0.046)  |                       |                         |                       |                       |                         |                       |
|               | ≥ 6 days         | 0.074***<br>(0.019)     | -0.456***<br>(0.048)  |                       |                         |                       |                       |                         |                       |
| Pay norm      |                  | 0.002<br>(0.002)        | 0.049***<br>(0.009)   |                       | 0.012*<br>(0.005)       | 0.031**<br>(0.009)    |                       | 0.007**<br>(0.003)      | 0.015<br>(0.016)      |
| Cainiao       |                  | 0.007<br>(0.016)        | -0.002<br>(0.014)     |                       | -0.046<br>(0.029)       | -0.020<br>(0.018)     |                       | -0.009<br>(0.003)       | 0.018<br>(0.039)      |
| $\rho$        |                  | 0.000<br>(0.055)        |                       |                       |                         | -0.005<br>(0.212)     |                       |                         | -0.022<br>(0.479)     |
| Controls      | Seller, Carrier  | Yes                     | Yes                   | Yes                   | Yes                     | Yes                   | Yes                   | Yes                     | Yes                   |
|               | Week, Day, Hour  | Yes                     | Yes                   | Yes                   | Yes                     | Yes                   | Yes                   | Yes                     | Yes                   |
|               | Holidays         | Yes                     | Yes                   | Yes                   | Yes                     | Yes                   | Yes                   | Yes                     | Yes                   |
| <i>N</i>      |                  | 14,546,267              | 5,507,349             |                       | 1,167,622               | 440,914               |                       | 134,374                 | 50,761                |
| LL            |                  | -12,113,335.0           |                       |                       | -909,463.4              |                       |                       | -102,589.3              |                       |

\*p&lt;0.05; \*\*p&lt;0.01; \*\*\*p&lt;0.001

Standard errors clustered at seller \* carrier level

The regression results and marginal effects provide sufficient empirical evidence on the significant relationship between delivery performance and logistics rating. To support the causal claims of our model, we conducted extensive analysis to address issues such as endogeneity and performed robustness checks. We report these results in §7.1 of the paper and the online appendix. The results and insights of this analysis strongly support the causal inference drawn from our model.

**Table 4** Average marginal effects of delivery time

| Model               | Response                              | No-promise speed          |                           |                           |                             | Two-day promise speed     |                             | One-day promise speed |                        |
|---------------------|---------------------------------------|---------------------------|---------------------------|---------------------------|-----------------------------|---------------------------|-----------------------------|-----------------------|------------------------|
|                     |                                       | 3 days<br>→<br>1/2 day(s) | 4 days<br>→<br>1/2 day(s) | 5 days<br>→<br>1/2 day(s) | ≥ 6 days<br>→<br>1/2 day(s) | 3 days<br>→<br>1/2 day(s) | ≥ 4 days<br>→<br>1/2 day(s) | 2 days<br>→<br>1 day  | ≥ 3 days<br>→<br>1 day |
|                     |                                       | (1)<br>(2)                | (3)                       | (4)                       | (5)                         | (6)                       | (7)                         | (8)                   | (9)                    |
| Probit<br>Selection | Pr( $\mathbf{1}(\text{Rating} > 0)$ ) | <b>-0.0094</b>            | <b>-0.0196</b>            | <b>-0.0269</b>            | <b>-0.0150</b>              | 0.0053                    | 0.0306                      | 0.0194                | 0.0139                 |
|                     | Pr(Rating = 1)                        | <b>-0.0026</b>            | <b>-0.0054</b>            | <b>-0.0091</b>            | <b>-0.0182</b>              | <b>-0.0144</b>            | <b>-0.0330</b>              | <b>-0.0125</b>        | <b>-0.0125</b>         |
| Ordered             | Pr(Rating = 2)                        | <b>-0.0008</b>            | <b>-0.0017</b>            | <b>-0.0027</b>            | <b>-0.0052</b>              | <b>-0.0032</b>            | <b>-0.0067</b>              | <b>-0.0021</b>        | <b>-0.0021</b>         |
| Probit              | Pr(Rating = 3)                        | <b>-0.0032</b>            | <b>-0.0064</b>            | <b>-0.0103</b>            | <b>-0.0190</b>              | <b>-0.0120</b>            | <b>-0.0240</b>              | <b>-0.0090</b>        | <b>-0.0090</b>         |
|                     | Pr(Rating = 4)                        | <b>-0.0069</b>            | <b>-0.0136</b>            | <b>-0.0214</b>            | <b>-0.0375</b>              | <b>-0.0307</b>            | <b>-0.0570</b>              | <b>-0.0262</b>        | <b>-0.0262</b>         |
|                     | Pr(Rating = 5)                        | <b>0.0135</b>             | <b>0.0271</b>             | <b>0.0435</b>             | <b>0.0799</b>               | <b>0.0603</b>             | <b>0.1207</b>               | <b>0.0498</b>         | <b>0.0498</b>          |

Bold values indicate significance less than 1%

## 5. Impact of Logistics Rating on Customer Purchase Probability

In this section, we analyze the second phase of our main research question: What is the impact of improved logistics rating for a seller on the customer purchase probability and the sales of third-party sellers on the platform? (Q1b).

We utilize the choice modeling framework to examine the impact of logistics ratings. In the traditional choice modeling literature (Train 2009), customers face the purchase decision of a product from a finite number of channels ( $C$ ). The customer with an intent to purchase compares the product from each channel based on attributes such as price, quality, channel preference, and additional variables. The customer receives the product's indirect utility, depending on the value and weight attached to each attribute. For example, a budget-constrained customer receives higher utility for a lower-priced product and a lower utility for a higher price. The net utility is assumed to be a weighted linear function of all product attributes and unobservable factors. The theory of utility-maximizing behavior predicts that the customer compares the product's net utility from different channels and makes a decision to purchase from one of the channels or not to purchase at all (typically referred to as the "outside option"). Each channel's choice probabilities or market shares derived under the standard MNL framework depend on the difference between the utilities. The outside option's utility is often set to a reference value of zero because the product is assumed to be unavailable for purchase beyond the  $C$  channels. This assumption does not hold in the context of our study.

We are constrained with the data on products, their attributes, and sales information only from sellers operating on a single platform, i.e., Tmall. We do not have data on similar products sold from sellers on other competing online platforms (e.g., JD.com) or brick-and-mortar stores. The possibility that customers can access similar products from these other channels exists. As a result, customers can receive a non-zero utility of purchase from other channels violating the assumption of zero utility for the outside option in the standard MNL model. We utilize the framework in Newman et al. (2014), which proposes an efficient and consistent estimation of choice-based models when one alternative is completely censored. Other channels of purchase, such as competing online platforms and brick-and-mortar stores, are entirely censored in our context. We build on this choice model and extend it to incorporate information on actual market size through the number of unique customer visits to the platform and allow utility specification of the “no-purchase” alternative to vary by time, which has not been considered in the prior literature.

We now develop a generic choice model for our context for an item sold by multiple sellers on the platform. Consider an item “ $j$ ” sold by a finite number of sellers ( $n_s$ ) on the Tmall platform. We assume that customers randomly arrive at the platform following a Poisson process with an arrival rate of  $\lambda$ . A customer arriving to the platform is confronted with a choice to purchase the item from one of the  $n_s$  sellers, or purchase from other online platforms, or brick-and-mortar stores, or to not purchase from any of these channels. For modeling convenience, we bundle the following choice options: purchasing the item from other online or physical channels and the no purchase option into a single category and refer it as a “no-purchase” alternative. Hence, the mutually exclusive and completely exhaustive alternatives comprise purchasing from one of the  $n_s$  sellers and a “no-purchase” alternative. For an arriving customer, the utility of purchasing item “ $j$ ” from seller “ $i$ ” on day “ $t$ ” is specified as

$$V_{ijt} = \alpha_{ij} + \mathbf{X}_{ijt} \cdot \boldsymbol{\beta} + \epsilon_{ijt} \quad i \in 1, 2, 3, \dots, n_s$$

$\mathbf{X}_{ijt}$  is a vector of different covariates of the item sold by seller “ $i$ ”. In our context,  $\mathbf{X}_{ijt}$  includes both time-varying covariates such as the average price of the item on day “ $t$ ”, cumulative average logistics rating of seller “ $i$ ” until day “ $t$ ”, average order quality of the item from seller “ $i$ ” on day “ $t$ ” and other covariates such as day of the week, week of the month, and holidays. The parameter  $\alpha_{ij}$  in the utility specification is a constant that captures customer’s preference to the item sold by the seller “ $i$ ”. The parameter vector  $\boldsymbol{\beta}$  captures the effect of each covariate on the utility specification. The idiosyncratic component  $\epsilon_{ijt}$  captures the random customer utility component that varies between all the  $n_s$  alternatives. We consider the utility from “no-purchase” alternative as

$$V_{0jt} = \gamma + \mathbf{Z}_{0jt} \cdot \boldsymbol{\delta} + \epsilon_{0jt}$$

The parameter  $\gamma$  is a non-zero constant that captures the average utility of “no-purchase” alternative. Newman et al. (2014) consider only a constant  $\gamma$  in the above specification of  $V_{0jt}$ . We allow the utility specification of “no-purchase” alternative to vary across the day of the week, week of the month, and holidays (Perdikaki et al. 2012, Lu et al. 2013, Fisher et al. 2018).  $\mathbf{Z}_{0jt}$  includes all these time covariates. The idiosyncratic component  $\epsilon_{0jt}$  captures random customer utility that varies between all the unobserved alternatives. For convenience, we remove the subscripts “ $j$ ” and “ $t$ ” from the notation and develop purchase probability expressions. Under the assumption of an independent and identical Gumbel distribution for each of the idiosyncratic components (McFadden 1984), the expression for the choice probability of purchasing the item from seller “ $i$ ” and no-purchase alternative are given by

$$P_i(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}) = \frac{e^{\alpha_i + \mathbf{X}_i \boldsymbol{\beta}}}{\sum_{i=1}^{n_s} (e^{\alpha_i + \mathbf{X}_i \boldsymbol{\beta}}) + e^{\gamma + \mathbf{Z}_0 \boldsymbol{\delta}}} \quad ; \quad P_0(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}) = \frac{e^{\gamma + \mathbf{Z}_0 \boldsymbol{\delta}}}{\sum_{i=1}^{n_s} (e^{\alpha_i + \mathbf{X}_i \boldsymbol{\beta}}) + e^{\gamma + \mathbf{Z}_0 \boldsymbol{\delta}}}$$

The probability of purchasing the item from some seller on the platform is given by

$$P(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}) = \sum_{i=1}^{n_s} P_i(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}) = 1 - P_0(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta})$$

The choice probabilities are function of the parameters  $\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}$  where  $\boldsymbol{\alpha} = [\alpha_i] \ i \in 1, 2, 3, \dots, n_s$ . Similar to Newman et al. (2014), we consider the choice probability of purchase from seller “ $i$ ” conditional that a purchase from one of  $n_s$  sellers is observed as

$$P_{i\otimes}(\boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{e^{\alpha_i + \mathbf{X}_i \boldsymbol{\beta}}}{\sum_{i=1}^{n_s} (e^{\alpha_i + \mathbf{X}_i \boldsymbol{\beta}})}$$

Our data provides information on daily unique number of customer visits for each item ( $N$ ) to the platform. The customer order data has a unique customer identifier for each purchase order and clicks, which allows us to compute the number of unique customer visits for each item in our data set. Let  $m_i, m$  represent the number of unique customer purchases of the item from seller “ $i$ ” and total number of customer purchases respectively, where  $m = \sum_{i=1}^{n_s} m_i$ . Using this information, we specify the likelihood function as follows

$$L(\lambda, \boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}) = \left( \frac{e^{-\lambda} \lambda^N}{N!} \right) \left( \frac{N!}{m!(N-m)!} P(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta})^m P_0(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta})^{N-m} \right) \left( \frac{m!}{m_1! m_2! \dots m_{n_s}!} \prod_{i=1}^{n_s} P_{i\otimes}(\boldsymbol{\alpha}, \boldsymbol{\beta})^{m_i} \right) \quad (3)$$

The likelihood function has three components. The first component refers to the probability that  $N$  unique customers visit the item page on the platform. The second component refers to the probability that  $m$  (out of  $N$ ) customers purchase on the platform. Using independence in the sequence of  $N$  customer arrivals, the number of observed customer purchases follows a Binomial distribution. The final part of the likelihood function captures the probability that  $m_i$  customers purchase the item from seller “ $i$ ” conditional that  $m$  purchases are observed on the platform. The parameters  $\lambda, \boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma, \boldsymbol{\delta}$  are

estimated using the data through the maximum likelihood estimation approach. The log-likelihood function excluding the components which are not a function of parameters is given by

$$LL(\lambda, \alpha, \beta, \gamma, \delta) = \left( -\lambda + N \cdot \log \lambda \right) + \left( m \cdot \log P(\alpha, \beta, \gamma, \delta) + (N - m) \cdot \log P_0(\alpha, \beta, \gamma, \delta) \right) + \left( \sum_{i=1}^{n_s} m_i \cdot \log P_{i\otimes}(\alpha, \beta) \right) \quad (4)$$

Maximizing the entire log-likelihood function is computationally intense. We utilize the two-step approach previously discussed in the literature (Pagan 1986, Newman et al. 2014) to estimate the parameters. In the first step, we maximize the right-end part of the log-likelihood function,  $LL_1(\alpha, \beta) = \sum_{i=1}^{n_s} m_i \cdot \log P_{i\otimes}(\alpha, \beta)$ .  $LL_1$  is a simple Multi-nomial Logit (MNL) model of choice of the item among different sellers operating on the platform conditional that a purchase is made. The “no-purchase” alternative is excluded from  $LL_1$ . Maximizing the log-likelihood function of MNL model that includes only a subset of exhaustive choices produces consistent parameter estimates (McFadden 1984). This implies that maximizing  $LL_1$  which does not include “no-purchase” alternative generates consistent parameter estimates for  $\alpha, \beta$ . It can be shown that  $LL_1$  is globally concave and has a unique maximum. Further, maximizing  $LL_1(\alpha, \beta)$  does not allow to estimate all  $\alpha_i$  ( $i \in 1, 2, 3, \dots, n_s$ ) due to identification problems. To estimate  $\gamma$  in the “no-purchase” alternative specification, one of the  $\alpha_i$  has to be set to 0. Matching shares property (Ben-Akiva et al. 1985, Ferguson et al. 2012) of a limited subset of discrete choice models such as MNL allows unique identifiable estimates of  $\alpha_i$  with one of them set to reference and by matching sample shares with purchase probabilities. Relying on this property, we set one observed alternative as the reference. Later in the second step, the utility of the “no-purchase” alternative is estimated against this reference. Let the estimates of  $\alpha, \beta$  after the first step estimation be  $\hat{\alpha}, \hat{\beta}$ . In the second step, we estimate parameters  $\lambda, \gamma, \delta$  by maximizing the remaining part of the log-likelihood function  $LL_2(\lambda, \gamma, \delta) = -\lambda + N \cdot \log \lambda + m \cdot \log P(\hat{\alpha}, \hat{\beta}, \gamma, \delta) + (N - m) \cdot \log P_0(\hat{\alpha}, \hat{\beta}, \gamma, \delta)$ . Note that  $\alpha, \beta$  are replaced with  $\hat{\alpha}, \hat{\beta}$  in  $LL_2$ . The estimated values after the second step are  $\hat{\lambda}, \hat{\gamma}, \hat{\delta}$ .

The platform updates the logistics rating at the seller level rather than for each item individually. The distribution of logistics rating that the seller accumulates at any point in time is reflected as the same for every item sold by the seller. As a result, we run the estimation for each item individually to examine the effect of logistics rating on customer purchase probability. For illustration, we choose items of a seller identified as “358” in our data. Seller 358 sold a total of 361 items, is among the bottom five sellers with the least logistics rating on the platform, and handled the highest volume of orders during April 2017. Table 5 lists the estimation results for three items of seller 358. Across the items, we find that the coefficient of average unit price is negative and statistically significant. This result implies that higher the item price, less is the utility for the customer to purchase the item, which is a trivial result. We find that the coefficient of average logistics rating is positive and

statistically significant, implying that higher the average logistics rating of a seller, higher is the utility for the customer to purchase the item from that seller. The constant  $\gamma$  is statistically significant and captures the average utility of the “no-purchase” alternative compared to an alternative whose  $\alpha_i$  is set to 0. We test the robustness of the causal link between this seller’s logistics rating and their sales for the three items designated in Table 5 by analyzing the instrument free reduced-form models using Gaussian copula correction (Park and Gupta 2012) in §7.2 of the paper.

**Table 5** Impact of logistics rating on customer purchase probability

| Variable                     | Item<br>“220636”<br>(1) | Item<br>“258478”<br>(2) | Item<br>“183163”<br>(3) |
|------------------------------|-------------------------|-------------------------|-------------------------|
| Average unit price           | −0.446**<br>(0.114)     | −0.376*<br>(0.135)      | −0.159*<br>(0.068)      |
| Average logistics rating     | 102.887***<br>(25.313)  | 50.088**<br>(15.974)    | 147.154*<br>(48.764)    |
| Average order quality rating | −0.851<br>(2.261)       | −0.046<br>(2.085)       | −12.990<br>(8.261)      |
| $\gamma$                     | 474.328**<br>(119.824)  | 233.915**<br>(79.097)   | 627.906*<br>(211.737)   |
| Week, Day                    | Yes                     | Yes                     | Yes                     |
| Holidays                     | Yes                     | Yes                     | Yes                     |
| Observations                 | 30                      | 30                      | 22                      |
| <i>LL</i>                    | −4.986                  | −1.764                  | −7.768                  |

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01

We extended this estimation approach to all the 361 items sold by this seller. We found that the effect of the average logistics rating is significant for 176 items ( $\approx 49\%$ ) and insignificant for the remaining 185 items. To arrive at more comprehensive insights across all the sellers operating on the platform, we extended our choice modeling estimation analysis for all the 119,276 items sold by 335 sellers.<sup>3</sup> We found that the effect of average logistics rating on customer purchase probability is significant for  $\approx 51\%$  of the items and insignificant for the remaining 49%. We conducted a *t*-test to examine if the price and volume sold vary significantly between items with and without the significance of average logistics rating on customer purchase probability. We found that the average unit price is not significantly different between the two groups. In contrast, the average volume is significantly higher for items where logistics ratings have a significant impact. Based on our extensive analysis, we conclude that the impact of logistics rating on sales is statistically significant for close to half of the items on the platform. Further, the improvement in logistics rating is potentially more beneficial for high volume items.

In summary, the results and insights from the analysis of Q1a and Q1b provide empirical evidence to the mechanism that connects delivery performance to sales through logistics ratings.

<sup>3</sup> We excluded ten sellers as the sales of their items are observed for less than five days in April 2017.

## 6. Logistics Performance and Sales

In this section, we examine our main research question: What is the impact of reducing delivery time on sales for a seller on the e-commerce platform? (Q1). We combine the results of our analysis from §4 and §5 to analyze the following specific question: What is the potential improvement in daily sales if a seller could potentially deliver all their three-day delivered orders in two days (i.e., reduce delivery time by one day for all their current three-day delivered orders)?

For illustration, we first focus on three-day delivered orders and seller 358 for two reasons. First, the number of three-day delivered orders is the largest in volume (makeup  $\approx 27.1\%$ ) for seller 358 (this is generally true across all the other sellers). Second, seller 358 is among the bottom five sellers with the least logistics rating on the platform and handled the highest number of orders. Hence, reducing delivery time by one day for three-day delivered orders is likely to impact sales of seller 358. We first quantify the impact of shipping all three-day delivered orders in two days on the seller's distribution and expected logistics rating. Next, we estimate the increase in customer purchase probability and sales due to the change in expected logistics rating.

During April 2017, seller 358 handled 353,299 orders with all orders shipped as no-promise speed orders. Of the total volume, there are no ratings available for 204,184 orders (57.79%). For the remaining 149,115 orders (42.21%), the conditional distribution of ratings is as follows: Rating "1" (count 4,348 ; proportion 2.92%), Rating "2" (1,342 ; 0.90%), Rating "3" (5,417 ; 3.63%), Rating "4" (9,530 ; 6.39%) and Rating "5" (128,478 ; 86.16%). We estimate the baseline expected rating to be 4.7198. Of the total 353,299 orders, the number of three-day delivered orders are 95,701.

Utilizing the marginal effects in Table 4, we tabulate change in the distribution and expected logistics rating for this seller due to shipping their three-day delivered orders in two days. Reducing the delivery time of the three-day delivered orders changes the likelihood of posting a rating and the value of rating for these customers' orders while others remain unaffected. As a result, we first separate the volume of orders into two categories: three-day and  $\neq$  three-day delivered orders. Table 6 provides a detailed enumeration of change in the distribution of ratings. Column (1) lists the category of orders by delivery time. Column (2) lists the proportion of availability of ratings for each category. For example, 57.89% of all three-day delivered orders have no rating, while the remaining 42.11% of orders have a rating. Columns (4) and (5) list the count and conditional distribution of ratings when the rating is available. Columns (6) and (7) list the marginal effects. Table 4 shows that delivering a three-day order in two days decreases (increases) the average probability of a customer posting a rating (no rating) by 0.0094. The probability of a customer posting a rating of 1, 2, 3, 4 decreases by 0.0026, 0.0008, 0.0032, and 0.0069 respectively, while that of rating 5 increases by 0.0135. Note that the marginal effects for  $\neq$  three-day delivered orders are 0 as these orders remain unaffected. Using the new distribution of ratings, we estimate the expected logistics rating to be 4.7265. We estimate

that shipping all the three-day delivered orders ( $\approx 27.1\%$  of its total orders) of seller 358 in two days results in an increase in expected logistics rating from 4.7198 to 4.7265.

**Table 6** Impact of shipping three-day delivered orders in two days on the distribution of logistics rating

| Delivery time | Baseline      |            |         |                            | Marginal Effects |                 | Improved      |                            |                 |
|---------------|---------------|------------|---------|----------------------------|------------------|-----------------|---------------|----------------------------|-----------------|
|               | % Ratings (s) | Rating (r) | Count   | % Rating (p <sub>r</sub> ) | Δs               | Δp <sub>r</sub> | % Ratings (s) | % Rating (p <sub>r</sub> ) | Expected Rating |
| (1)           | (2)           | (3)        | (4)     | (5)                        | (6)              | (7)             | (8)           | (9)                        | (10)            |
| = three days  | 57.89%        | -          | 55,399  | -                          | 0.94%            |                 | 58.83%        | -                          |                 |
|               |               | 1          | 979     | 2.43%                      |                  |                 |               | 2.17%                      |                 |
|               |               | 2          | 295     | 0.73%                      |                  |                 |               | 0.65%                      |                 |
|               | 42.11%        | 3          | 1,278   | 3.17%                      | −0.94%           | −0.32%          | 41.17%        | 2.85%                      |                 |
|               |               | 4          | 2,553   | 6.33%                      |                  | −0.69%          |               | 5.64%                      |                 |
|               |               | 5          | 35,197  | 87.33%                     |                  | 1.35%           |               | 88.68%                     |                 |
| ≠ three days  | 57.76%        | -          | 148,785 | -                          | 0.00%            |                 | 57.76%        | -                          |                 |
|               |               | 1          | 3,369   | 3.10%                      |                  | 0.00%           |               | 3.10%                      |                 |
|               |               | 2          | 1,047   | 0.96%                      |                  | 0.00%           |               | 0.96%                      |                 |
|               | 42.24%        | 3          | 4,139   | 3.80%                      | 0.00%            | 0.00%           | 42.24%        | 3.80%                      |                 |
|               |               | 4          | 6,977   | 6.41%                      |                  | 0.00%           |               | 6.41%                      |                 |
|               |               | 5          | 93,281  | 85.73%                     |                  | 0.00%           |               | 85.73%                     |                 |
| 4.7265        |               |            |         |                            |                  |                 |               |                            |                 |

We now estimate the increase in customer purchase probability and sales due to the change in expected logistics rating. We provide the general expressions for the increment in sales for a seller “ $i$ ”. Let  $j, n_i$  represent the item identifier and the number of items sold by seller “ $i$ ” where  $j \in 1, 2, 3, \dots, n_i$ . The predicted sales of item “ $j$ ” of seller “ $i$ ” on day “ $t$ ” under baseline expected logistics rating is given by  $S_{ijt1} = pr_{ijt1} \cdot N_{ijt} \cdot p_{ijt}$ , where  $pr_{ijt1}$ ,  $N_{ijt}$ ,  $p_{ijt}$  denote purchase probability, unique visitors, and average price of item “ $j$ ” respectively on the Tmall platform on day “ $t$ ” under current baseline logistics performance. Similarly, the predicted sales of the same item under improved logistics rating by reducing delivery time from three days to two days is given by  $S_{ijt2} = pr_{ijt2} \cdot N_{ijt} \cdot p_{ijt}$ , where  $pr_{ijt2}$ ,  $N_{ijt}$ ,  $p_{ijt}$  denote purchase probability, unique visitors, and average price of item “ $j$ ” respectively on the Tmall platform on day “ $t$ ” under improved logistics performance. The difference,  $pr_{ijt2} - pr_{ijt1}$  represents the incremental customer purchase probability resulting from the change from baseline to improved logistics rating performance. The average daily sales for seller “ $i$ ” from all their  $n_i$  items, over “ $D$ ” days under baseline and improved logistics rating are given by  $S_{i1} = \frac{1}{D} \sum_{t=1}^D \sum_{j=1}^{n_i} S_{ijt1}$ ,  $S_{i2} = \frac{1}{D} \sum_{t=1}^D \sum_{j=1}^{n_i} S_{ijt2}$  respectively. The percentage change is average daily sales for seller “ $i$ ” is given by  $(\frac{S_{i2}}{S_{i1}} - 1) \cdot 100\%$ . We implement the above computation and find that the improvement in average daily sales for the seller 358 to be 10.3%. In conclusion, we find that reducing the delivery time of all the three-day delivered orders ( $\approx 27.1\%$  of its total orders) for the seller 358 to two days results in an increase of average daily sales for this seller by 10.3%.

We repeated the entire process of estimating the impact of shipping each seller’s three-day delivered orders in two days on sales to derive general insights for all sellers. We found that reducing the

delivery time of all three-day delivered orders (that makeup  $\approx 35\%$  of the total orders across all sellers) to two days improves the average daily seller sales by 13.3% on the platform.

To derive further insights, we summarize the results on the improvement in sales by item volume group categories and item price group categories. We first sort the sellers in the ascending order of % three-day volume orders handled by the seller. The top 25% and the bottom 25% of the sellers are labeled into “Low” and “High” % three-day volume group categories, respectively, while the remaining are labeled as “Medium” group category. Table 7 lists the average change in % sales by improving logistics performance for each % three-day volume group category. We find that reducing delivery time for all three-day delivered orders (that makeup  $\approx 28.9\%$  -  $39.2\%$  of the total orders) to two days improves the average daily sales by 9.0% - 22.5% on the platform. The impact is most significant for sellers in the “High” % three-day volume group category.

**Table 7 Improvement in sales for sellers grouped by % three-day volume**

| Seller group | % Three-day order volume<br>(Mean) | % Change in daily sales<br>(Mean) |
|--------------|------------------------------------|-----------------------------------|
| Low          | 28.9%                              | 9.0%                              |
| Medium       | 34.8%                              | 10.8%                             |
| High         | 39.2%                              | 22.5%                             |

Next, we summarize the results on the improvement in sales by improving logistics performance by item price group. We sort the items sold on the platform in the ascending order of prices. The top 25% and the bottom 25% of the items are labeled as “Low” and “High” price item group categories, respectively, while the remaining are labeled as “Medium” price groups. Table 8 lists the average change in % sales for each item price group. On average, we find that the improvement in sales by decreasing delivery time by one day is most prominent for the high priced item category.

**Table 8 Improvement in sales for items grouped by price**

| Item price group | Unit price (Yuan)<br>(Mean) | % Change in daily sales<br>(Mean) |
|------------------|-----------------------------|-----------------------------------|
| Low              | 6.44                        | 13.6%                             |
| Medium           | 26.59                       | 22.0%                             |
| High             | 287.78                      | 28.2%                             |

We now summarize the results on the improvement in sales for each seller-item combination by the item price and % three-day volume seller group. Table 9 lists the average change in % sales for each combination. We conclude that sales improvement is more significant for high priced items and sellers who manage a high volume of % three-day volume orders.

**Table 9 Improvement in sales by % three-day volume and item price group**

| Seller group | Item price group |        |       |
|--------------|------------------|--------|-------|
|              | Low              | Medium | High  |
| Low          | 6.8%             | 6.5%   | 3.6%  |
| Medium       | 5.9%             | 5.5%   | 23.9% |
| High         | 20.4%            | 53.2%  | 54.6% |

Lastly, we summarize the improvement in sales across different sellers grouped by average delivery time. We first sort the sellers in the ascending order of their average delivery time of orders. The top 25% and the bottom 25% of the sellers are labeled into the “Low” and “High” group categories, respectively, while the remaining are labeled as the “Medium” group category. Table 10 lists the average change in % sales by improving logistics performance for each group category. We find that the impact is most significant for sellers in the “Medium” group category as the sellers in this group observed more three-day delivery time orders.

**Table 10 Improvement in sales for sellers grouped by average delivery time**

| Seller group | Average delivery time (days)<br>(Mean) | % Three-day order volume<br>(Mean) | % Change in daily sales<br>(Mean) |
|--------------|----------------------------------------|------------------------------------|-----------------------------------|
| Low          | 2.97                                   | 35.0%                              | 8.8%                              |
| Medium       | 3.55                                   | 38.0%                              | 18.3%                             |
| High         | 4.12                                   | 28.0%                              | 7.7%                              |

## 7. Robustness Analysis

The first phase of the main research question (Q1) that we examine is: What is the impact of delivery time on the logistics rating provided by customers for sellers from whom they purchased the product? (Q1a). We employed the Heckman ordered probit regression model on each subset of promise speed orders separately to explain the variation in the customer-provided logistics service ratings using covariates such as delivery time, order amount paid, and other variables. The specification is potentially subject to endogeneity resulting from independent variables (Certo et al. 2016). Endogeneity emanating from independent variables can occur in multiple forms such as selection bias (selection on observables and unobservables), simultaneity, and omitted variable bias (Ho et al. 2017a). We analyze the robustness of our results due to selection on unobservables endogeneity in this section. The analysis and the results of remaining endogeneity issues - selection on observables, simultaneity, and omitted variable bias are reported in §A of the online appendix.

Similarly, the second phase of the main research question that we examine is: What is the impact of improved logistics rating on the customer purchase probability of a seller and, as a result, sales of a third-party seller on the platform? (Q1b). To analyze this question, we adopted a structural econometric modeling approach by imposing an analytical structure derived from the utility maximization

principle (Train 2009). Specifically, we utilized and extended the customer choice model framework in Newman et al. (2014), which proposes an efficient and consistent estimation of choice-based models when one alternative is completely censored. As elaborated in prior literature, choice models employed based on structural econometric modeling provide causal estimates of logistics ratings on customer purchase probability (Ho et al. 2017a). However, we test the robustness of the causal link between seller's logistics rating and their sales by analyzing instrument free reduced-form models using Gaussian copula correction (Park and Gupta 2012).

We first elaborate and present the analysis of selection on unobservables endogeneity for Q1a, followed by the analysis of Gaussian copula correction for Q1b.

### 7.1. Selection on Unobservables

The selection on unobservable problems arises when the unit's assignment to the treatment group is due to variables that are not observed and affect the outcome. The general quasi-experimental methods used to estimate the causal effect of treatment under the selection of unobservables problem are Regression Discontinuity Design (RDD) and Difference-In-Differences (DID) method (Angrist and Pischke 2008). We employ the RDD approach as DID requires observations during pre-treatment and post-treatment periods for both the treatment and control groups. Such kind of data generating process is not viable in our context. To employ RDD, we need to identify a threshold value of a continuous variable (also referred to as forcing variable) that differentiates between the treatment and control group observations. The observations around the threshold value of the forcing variable are mostly similar. The difference in output values between treatment and control observations accounting for any observable variables provides the causal estimate of the treatment effect.

In our context, we consider the time of order arrival as the forcing variable and the threshold value of 12 AM (midnight) that identifies the treatment and control group orders. We focus on the no-promise speed orders that the platform received in the one-hour interval before and after midnight. For example, consider the three-day delivered orders that arrived from 11:00 PM - 11:59 PM on the current day and the two-day delivered orders that arrived during 12 AM - 1 AM on the next day. Drawing parallels from the RDD framework, if the two-day delivered orders are considered in the control group, the three-day delivered orders can be considered in the treatment group. These orders around the threshold of 12 AM (midnight) are not significantly different, given the time instant orders were received after controlling for other observable covariates. Hence, the difference in logistics rating between these two-day and three-day delivered orders provides the causal estimate of the change in logistics rating due to an increase in an additional day of delivery time. The model specification is as follows. Let  $D_i$  be the indicator which takes 1 if the order is part of the treatment group, otherwise

0;  $D_i = \mathbf{1}[F_i < c]$ ,  $F_i$  is the time instant the order was received, and  $c$  is 12 AM. We run the following ordered probit model

$$\Pr(Y_i = r) = \Pr\left(K_{r-1} < (\alpha + \beta \cdot D_i + \gamma_1 \cdot (F_i - c) + \gamma_2 \cdot D_i \cdot (F_i - c) + \mathbf{X}_i \cdot \boldsymbol{\delta} + \epsilon_i) \leq K_r\right)$$

$Y_i$  is the ordinal outcome for the order  $i$ . The variable  $r$  takes values 1, 2, 3, 4 & 5.  $K_0$  to  $K_5$  are cut-off points on the underlying customer latent response curve with  $K_0 = -\infty$  and  $K_5 = \infty$ .  $\epsilon_i$  is the idiosyncratic random error term assumed to follow normal distribution. The coefficient of  $D_i$ ,  $\beta$  provides the causal effect. It is unnecessary to include other covariates in the regression, even if they are essential in the selection criterion. However, including available covariates can help reduce any small-sample bias (Imbens and Lemieux 2008).  $\mathbf{X}_i$  includes all observable covariates such as order pay, channel, seller, carrier, and time fixed effects.

Table 11 lists the RDD results. Column (1) lists the results of three-day delivered treatment group orders against two-day delivered control group orders. We find that the coefficient of  $D$  in column (1) is negative and statistically significant. This result indicates that the logistics rating provided by three-day delivered no-promise speed customers is lower than two-day delivered no-promise speed customers. Similarly, in columns (2) and (3), we compare three-day vs. four-day delivered orders and four-day vs. five-day delivered orders received around one-hour intervals from midnight. To conclude, we note that if the endogeneity in our data generating process is due to selection on unobservables, our results from RDD provide the causal effects of delivery time on logistics rating for no-promise speed customer orders.

The RDD model, although causal, generates a local estimate due to the selection of no-promise orders observed exclusively around midnight, and, hence, its results cannot be generalized to the entire data set. Hence, to compare the main model (Heckman ordered probit) and RDD estimates (Table 11), we analyze the Heckman ordered probit model on the limited number of observations utilized for the RDD model. Table 12 summarizes the comparison. Column (2) lists the results of the Heckman ordered probit model employed on the RDD data with two-day delivery time orders (control group) vs. three-day delivery time orders (treatment group). Column (3) captures the RDD estimates (column (1) from Table 11). A comparison of columns (2) and (3) shows that the three-day delivery time coefficient is significant and similar in magnitude for both models. Similarly, columns (4) and (5) compare the estimates of the two models for three-day vs. four-day delivery time orders. We find the estimates from the causal RDD method and the Heckman model are very close in all cases, and the magnitude of the bias of the Heckman model is less than 5%. This eases any causal concerns of the delivery time effect on logistics rating for no-promise speed orders in the Heckman model.

**Table 11** RDD results : one-hour window around 12 AM (midnight)

| Control<br>Treatment | 2-day<br>3-day        | 3-day<br>4-day         | 4-day<br>5-day        |
|----------------------|-----------------------|------------------------|-----------------------|
| Variable             | (1)                   | (2)                    | (3)                   |
| $D$                  | -0.1560*<br>(0.0673)  | -0.1760**<br>(0.0577)  | -0.1160<br>(0.0834)   |
| $F - c$              | -0.0008<br>(0.0006)   | -0.0017***<br>(0.0004) | -0.0014*<br>(0.0006)  |
| $D \cdot (F - c)$    | -0.0003<br>(0.0007)   | -0.0014**<br>(0.0005)  | -0.0013<br>(0.0007)   |
| Pay                  | 0.0000***<br>(0.0000) | 0.0001***<br>(0.0000)  | 0.0002***<br>(0.0000) |
| Cainiao              | 0.0653***<br>(0.0132) | 0.0289*<br>(0.0121)    | -0.0339<br>(0.0222)   |
| Seller, Carrier      | Yes                   | Yes                    | Yes                   |
| Week, Day, Hour      | Yes                   | Yes                    | Yes                   |
| Holidays             | Yes                   | Yes                    | Yes                   |
| $N$                  | 83,172                | 166,047                | 74,972                |
| LL                   | -33,090.3             | -74,125.3              | -37,144.7             |

\*p&lt;0.05; \*\*p&lt;0.01; \*\*\*p&lt;0.001

Clustered standard errors at seller \* carrier level

**Table 12** Ordered Regression and Regression Discontinuity (RDD) results : No-promise speed orders

| Variable          | 2 days (Control) vs 3 days (Treatment) |                    | 3 days vs 4 days    |                     | 4 days vs 5 days  |                   |
|-------------------|----------------------------------------|--------------------|---------------------|---------------------|-------------------|-------------------|
|                   | Ordered Probit                         | RDD                | Ordered Probit      | RDD                 | Ordered Probit    | RDD               |
| (1)               | (2)                                    | (3)                | (4)                 | (5)                 | (6)               | (7)               |
| 3 days            | -0.149*<br>(0.065)                     | -0.156*<br>(0.067) |                     |                     |                   |                   |
| 4 days            |                                        |                    | -0.163**<br>(0.058) | -0.176**<br>(0.058) |                   |                   |
| 5 days            |                                        |                    |                     |                     | -0.101<br>(0.083) | -0.116<br>(0.083) |
| Pay norm, Cainiao | Yes                                    | Yes                | Yes                 | Yes                 | Yes               | Yes               |
| Seller, Carrier   | Yes                                    | Yes                | Yes                 | Yes                 | Yes               | Yes               |
| Week, Day, Hour   | Yes                                    | Yes                | Yes                 | Yes                 | Yes               | Yes               |
| Holidays          | Yes                                    | Yes                | Yes                 | Yes                 | Yes               | Yes               |
| $N$               | 83,172                                 | 83,172             | 166,047             | 166,047             | 74,972            | 74,972            |
| LL                | -33,090.3                              | -33,090.2          | -74,129.5           | -74,125.3           | -37,146.4         | -37,144.7         |

\*p&lt;0.05; \*\*p&lt;0.01; \*\*\*p&lt;0.001

Standard errors clustered at seller \* carrier level

## 7.2. Instrument-Free Reduced Form Model Using Gaussian Copula Correction

We analyze the instrument free reduced-form models using Gaussian copula correction (Park and Gupta 2012) to strengthen the evidence of the causal link between seller's logistics rating and seller sales. The instrument-free linear model is estimated considering the seller's average logistics rating as the endogenous variable. We employ the model for three items of seller "358" designated in Table 5. The results of the instrument-free reduced-form models are listed in Table 13. The dependent variable

is the number of customer purchases for the chosen item. We find a statistically significant positive coefficient for average logistics rating for all three items, suggesting that the number of customer purchases for the chosen item increases with the seller's average logistics rating. The coefficient estimates from the Gaussian copula correction and choice model (Table 5) differ as the dependent variable is different in both the models. In the Gaussian copula correction model, the dependent variable is the number of customer purchases for the chosen item. In contrast, the choice model's dependent variable is the customer's utility to purchase the item or customer purchase probability. In summary, these results are directionally consistent with the choice model results and strengthen the evidence of the causal link between seller's logistics rating to their sales.

**Table 13**      **Gaussian Copula Correction : Impact of logistics ratings on item sales of seller 358**

| Variable                 | Dependent Variable : Number of Customer Purchases<br>(Endogeneous Average logistics rating) |                          |                          |
|--------------------------|---------------------------------------------------------------------------------------------|--------------------------|--------------------------|
|                          | Item<br>"220636"                                                                            | Item<br>"258478"         | Item<br>"183163"         |
|                          | (1)                                                                                         | (2)                      | (3)                      |
| Average unit price       | -7.809**<br>(2.427)                                                                         | -9.936**<br>(3.593)      | -3.148***<br>(1.092)     |
| Average logistics rating | 2,468.154***<br>(27.110)                                                                    | 9,531.651***<br>(27.413) | 1,111.482***<br>(15.953) |
| Week, Day                | Yes                                                                                         | Yes                      | Yes                      |
| Holidays                 | Yes                                                                                         | Yes                      | Yes                      |
| Observations             | 30                                                                                          | 30                       | 22                       |
| LL                       | 103.2                                                                                       | 129.5                    | 81.6                     |

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01

A caveat we wish to stress is that the instrument-free reduced form models using Gaussian copula correction discard two components essential for empirically linking seller's logistics rating and seller sales - (i) Sellers or markets outside the Tmall platform, (ii) Information on actual market size through the number of unique customer visits to the platform. These two components, largely ignored in the prior literature, are incorporated in our choice model. Hence we retain our original choice model as the main model in the paper while providing the reduced-form Gaussian copula correction as a robustness check.

## 8. Conclusion & Managerial Implications

Online Retail is one of the fast-growing business in e-commerce. Online retailers provide several advantages: avoiding travel to physical stores, accessing a wide variation of products via a digital channel, smooth one-click purchase transaction, and many more. Customer experience plays a cardinal role in creating a monetary transaction for sellers on the platform. Increasing customer traffic to the platform is the key to the online retail business's survival and growth. These platforms can

increase customer traffic either by investing efforts to retain and make the existing customer shop more or attract new customers through word-of-mouth (WOM). A means to achieve this for sellers is by selling products with lower prices, superior quality, and provide enriching service experience compared to their competitors. With limited variations in prices and product quality across competing platforms, service experience can be a differentiating factor in attaining a market share. As a result, online retailers typically provide premium membership, which promises fast delivery services such as same-day or next-day deliveries, hoping that customers are impressed by an excellent service experience, provide useful feedback to the platform, and return to shop in the future. Faster deliveries, although convenient to customers, results in high costs to online retailers. The additional costs range from starting new fulfillment centers to air-cargo operations. As a result, online retailers need to examine if the investment in fast shipping infrastructure has a payoff in increasing sales.

The mechanism of improved financial outcomes from superior logistics performance can be a result of a customer's response to: (i) their own recent or past service experience, or (ii) to accumulated feedback from previous customers' experience on the common platform (Word-of-Mouth). This paper focuses on the second mechanism (word-of-mouth effect through logistics ratings) by providing an empirical validation of this mechanism. We analyze the mechanism by examining the following two research questions individually. First, what is the impact of delivery time on the logistics rating provided by a customer? (Q1a). Second, what is the impact of improved logistics ratings on customer purchase probability and sales for the seller? (Q1b). We then combine our results from Q1a and Q1b to answer our main research question Q1.

Next, we summarize the principal results and insights from our study and provide relevant managerial recommendations. First, our study finds that reducing the delivery time of all three-day delivered orders (that makeup  $\approx 35\%$  of the total orders) to two days improves the average daily sales by 13.3% on the platform. This result has an immense practical significance for online retailers, as it quantifies the potential benefit of investing in improved logistics performance. Besides, the result emphasizes that delivery time performance and logistics ratings, which measure service quality, are essential drivers of a customer's likelihood of purchase and, hence, sales on e-commerce platforms. Hence, we recommend that e-commerce platforms and sellers should pay attention to logistics performance quality, in addition to product quality in driving traffic and sales.

Second, we find that longer delivery time increases the likelihood of posting a logistics rating and results in lower logistics rating by non-priority customers (no-promise speed customers in our context) for delivery time beyond two days. This result is surprising because it suggests that while customers with no promise speed do not have an explicit delivery deadline, they seem to assume a two-day delivery deadline implicitly. Hence, they are disappointed by deliveries that take longer than two days. Thus, no-promise speed customers seem to have been conditioned to expect deliveries in

two days. Hence, e-commerce platforms should not take no-promise speed customers for granted and should pay attention to these customers' delivery performance.

Third, for priority customers (two-day and one-day promise speed customers), we find that longer delivery time results in lower logistics ratings for delivery time beyond their anticipated delivery date, but has no impact on their likelihood of posting a logistics rating. In summary, for any customer type, we conclude that shortening delivery time results in a higher rating conditional that the customer provides a rating. Because customers post a higher rating upon observing superior delivery performance, increasing the likelihood of customer posting a rating is a key to achieve a significant jump in logistics rating for the seller and on the platform. As a result, we suggest that the platform or sellers reach out to customers and encourage them to participate in the rating process, especially when they experience superior delivery performance.

Fourth, our results suggest that there is no additional incentive for delivering orders earlier than the anticipated delivery date for priority customers. For example, we find that delivering two-day promise speed customer orders in one day does not provide additional value either in the increased likelihood of posting a rating or a higher rating. As a result, if the goal is to improve logistics rating on the platform, we suggest that managers need to alter their shipment policy such that priority orders are delivered close to their anticipated delivery date and allocate any remaining capacity to delivering no-promise speed orders earlier. For example, consider a simple scenario where a seller receives ten no-promise speed and ten two-day promise orders on a given day. Let us assume that the seller knows that all the no-promise orders can be delivered to the customers in two days, and all the two-day promise orders can be delivered in one day. Assuming the seller has a capacity of shipping a maximum of ten orders per day, a policy of shipping all no-promise speed orders on the current day and two-day promise orders on the next day is better than a policy that ships all two-day orders before no-promise orders. This is because both no-promise speed and two-day promise speed orders will be delivered in two days with the first policy, which is better than delivering two-day promise speed orders in one day and no-promise speed orders in three days. This example suggests that always prioritizing orders with a promise speed over orders with no promise speed may be sub-optimal.

Fifth, our analysis provides a specific recommendation on which orders (three-day delivered orders) retailers should focus on for delivery time improvement. Our results show that sellers can achieve the most significant improvement in sales by reducing the delivery time of no-promise speed three-day delivered orders by one day, which is better than other choices, such as delivering four-day delivered orders in three days. This is because a large chunk of orders ( $> 90\%$ ) on the Tmall platform are no-promise speed orders, and the mode of the delivery time histogram for no-promise speed orders is three days ( $\approx 35\%$ ).

Lastly, our study also provides a strong methodological framework that managers can potentially use and implement to assess if the future increase in sales due to improved delivery performance can offset the increase in costs of investing in additional infrastructure for faster shipping. By quantifying the impact of delivery time performance on sales, our study provides a method to assess the benefits of faster deliveries. Our insights are relevant to independent sellers and e-commerce platform managers who aim to improve long-term online customer traffic and sales.

Like every research study, our study also has its limitations. For example, our study does not model the strategic interaction among the sellers on the platform. In other words; our study does not provide answers to questions such as - how should other sellers modify their delivery speed options when a focal seller provides a faster shipping speed. For example, should other sellers offer a “same day delivery” promise speed when the focal seller offers the same improved shipping speed option? These issues are of importance for future research but outside the scope of our current study due to limitations of our data set. In conclusion, our model and estimates apply to settings with limited strategic interaction between sellers operating on and across platforms. We leave the exploration of such further questions for future research.

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# Online Supplement to “Logistics Performance, Ratings, and its impact on Customer Purchasing Behavior and Sales in E-commerce Platforms”

In this online supplement, we provide details on several robustness checks and analysis on endogeneity issues.

## A. Endogeneity

The main research question we examine in the study is: What is the impact of reducing delivery time on sales for a third-party seller on the e-commerce platform? (Q1). We identify the mechanism behind the relationship between delivery performance and sales in two phases. In the first phase, we examine: What is the impact of reducing delivery time on the logistics rating provided by customers for sellers from whom they purchased the product? (Q1a). In the second phase, we examine: What is the impact of improved logistics ratings on customer purchase probability and sales for the seller? (Q1b).

To establish causality of the impact of delivery performance on sales, we must attain causal estimates from both the research questions, Q1a and Q1b. In Q1b, we addressed the endogeneity issue and tested the robustness of the causal link between seller's logistics rating and their sales by analyzing the instrument free reduced-form models using Gaussian copula correction (Park and Gupta 2012) in §7.2 of the main paper. In Q1a, the endogeneity resulting from independent variables in the Heckman ordered probit regression model can undermine the causal link between delivery performance and customer-provided logistics rating. The endogeneity emanating from independent variables can occur in multiple forms such as selection bias (selection on observables and unobservables), simultaneity, and omitted variable bias (Ho et al. 2017a). We analyzed the robustness of our results due to selection on unobservables endogeneity in §7.1 of the main paper. We report the analysis and the results of remaining endogeneity issues - selection on observables, simultaneity, and omitted variable bias in this online appendix.

As is well known in the literature, “Endogeneity is not a problem that can be solved” (Ketokivi and McIntosh 2017), and “there is no way to statistically ensure that an endogeneity problem has been solved” (Roberts and Whited 2013, p. 498). Establishing causality is extremely hard unless the study is conducted by rigorously designing a controlled experiment (Ketokivi and McIntosh 2017, Duflo and Banerjee 2017). Endogeneity can occur in any of the forms mentioned above in non-experimental observational data like our data in this study. Suppose the endogeneity is due to selection bias (selection on observables and unobservables). In that case, we can employ quasi-experimental methods such as Propensity Score Matching (Rosenbaum and Rubin 1983) and Regression Discontinuity Design (Angrist and Pischke 2008), drawing parallels to the quasi-experimental data structure in our context. On the contrary, if the endogeneity is due to simultaneity and omitted variable bias, we either attempt to rule out or minimize the extent to which these problems can cause bias in the estimates.

This online appendix begins with an analysis on endogeneity issues: the selection on observables, simultaneity, and omitted variable bias separately for Q1a and ends with examining the robustness of the Heckman ordered probit regression results for an alternate month (May 2017).

### A.1. Selection on Observables

Research studies which typically examine the impact of a policy on outcomes face self-selection bias problems. Some examples of studies include the impact of responsible scheduling practices on store profitability (Kesavan et al. 2020), and adopting programs such as ISO certification on operational and financial outcomes (Levine and Toffel 2010). Self-selection bias occurs when units are self-selected into the control or treatment group based on specific observable or unobservable attributes. If a unit's assignment to treatment is due to observable variables, then the problem is selection on observables. On the contrary, if the assignment is due to variables that are not observed, the result is a selection on unobservables. Investigating such policy effects using standard regression models can lead to biased and inconsistent causal estimates (Ho et al. 2017a).

Propensity Score Matching method is a standard approach used to address the selection on observables problem as it is easy to implement and does not suffer from the curse of dimensionality. In our context, the selection on observables problem can arise when sellers (or carriers) self-select specific customer orders for shipping based on observed variables such as order amount paid, promise speed, channel, and additional variables, which impact their delivery time. To fit the selection on observables framework into our study, we need to define what qualifies an order to be in the control or treatment group. We label all the orders which are delivered on-time (delivery time  $\leq$  promise speed) as part of the control group and delayed orders (delivery time  $>$  promise speed) in the treatment group. For one-day promise speed data, orders delivered in one day are assigned to the control group, and orders with a delivery time  $\geq$  two days are assigned to the treatment group. Similarly, for two-day promise speed data, orders delivered in one day or two days are assigned to the control group, and orders with a delivery time  $\geq$  three days are assigned to the treatment group. No-promise speed orders are considered to be on-time all the time by definition. As a result, we analyze the propensity score matching method only on two-day promise speed and one-day promise speed orders data.

Let  $D_i$  be the treatment indicator for each order  $i$  whose value is 1 for a delayed order and 0 otherwise. The potential outcomes (logistics rating) for delayed and on-time orders are  $Y_i(1), Y_i(0)$ , respectively. We first match each delayed order with the on-time orders using covariates such as order pay, channel, seller, carrier, and exhaustive time effects. Let  $M_m(i)$  be  $m$  matched orders for each order  $i$ . The expressions for observed or estimated outcomes is given by

$$Y_i(\hat{1}) = \begin{cases} Y_i & D_i = 1 \\ \frac{1}{m} \cdot \sum_{j \in M_m(i)} Y_j & D_i = 0 \end{cases} \quad ; \quad Y_i(\hat{0}) = \begin{cases} Y_i & D_i = 0 \\ \frac{1}{m} \cdot \sum_{j \in M_m(i)} Y_j & D_i = 1 \end{cases}$$

$Y_i$  is the observed logistics rating for order  $i$ .  $Y_i(\hat{0}), Y_i(\hat{1})$  are the observed or estimated logistics rating for on-time and delayed order. For a dataset with total  $n$  orders where  $p$  are on-time, and  $q$  are delayed, the Average Treatment Effect (ATE), Average Treatment Effect of Treated (ATT) and Average Treatment Effect of Untreated (ATU) are given by

$$ATE = \frac{1}{n} \cdot \sum_{i=1}^n (Y_i(1) - Y_i(0)) \quad ; \quad ATT = \frac{1}{q} \cdot \sum_{i=1}^q (Y_i(1) - Y_i(0)) \quad ; \quad ATU = \frac{1}{p} \cdot \sum_{i=1}^p (Y_i(1) - Y_i(0))$$

The variants of treatment effects listed above denote the average difference in logistics rating for delayed and on-time orders accounting for all other covariates. Tables 14 and 15 list results of the propensity score

method using nearest neighbors matching with 1, 3, 5, and 10 observations and caliper 0.1 on two-day promise speed and one-day promise data. Column (1) in both the tables provide a difference in means of observed covariates before matching. Columns (2) and (3) list the difference in means post matching and percentage reduction in bias with one nearest neighbor matching. Regardless of the nearest neighbor matching level, we find all variants of treatment effects (ATE, ATT, ATU) to be negative and statistically significant. This result suggests that the average logistics rating of orders whose delivery time is beyond their anticipated delivery date is significantly lower than that of orders delivered within their anticipated delivery date. To conclude, we note that if the endogeneity in our data generating process is due to selection on observables, our results from the propensity score matching method provide the causal effects of delivery time on logistics rating for two-day and one-day promise speed customer orders.

**Table 14 Propensity Score Matching results: Two-day promise speed**

| Variable  | Before Matching              |                              |                              | After Matching               |                              |                              |                              |                              |                              |
|-----------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|           | Difference<br>in mean<br>(1) | NN(1)                        |                              | NN(3)                        |                              | NN(5)                        |                              | NN(10)                       |                              |
|           |                              | Difference<br>in mean<br>(2) | % reduction<br> Bias <br>(3) | Difference<br>in mean<br>(4) | % reduction<br> Bias <br>(5) | Difference<br>in mean<br>(6) | % reduction<br> Bias <br>(7) | Difference<br>in mean<br>(8) | % reduction<br> Bias <br>(9) |
| Order Pay | 39.13***                     | -2.03                        | 94.8                         | 6.79                         | 82.7                         | -8.64                        | 77.9                         | 7.04                         | 96.3                         |
| Cainiao   | -0.0025                      | -0.0024                      | 6.3                          | -0.0044                      | -76.4                        | -0.0048                      | 87.9                         | -0.0040                      | -140.7                       |
| ATE       |                              | -0.165***                    |                              | -0.174***                    |                              | -0.171***                    |                              | -0.174***                    |                              |
| ATT       |                              | -0.170***                    |                              | -0.167***                    |                              | -0.167***                    |                              | -0.169***                    |                              |
| ATU       |                              | -0.165***                    |                              | -0.174***                    |                              | -0.172***                    |                              | -0.174***                    |                              |

Note : Matching performed at caliper 0.1

\*p<0.05; \*\*p<0.01; \*\*\*p<0.001

**Table 15 Propensity Score Matching results: One-day promise speed**

| Variable  | Before Matching              |                              |                              | After Matching               |                              |                              |                              |                              |                              |
|-----------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|           | Difference<br>in mean<br>(1) | NN(1)                        |                              | NN(3)                        |                              | NN(5)                        |                              | NN(10)                       |                              |
|           |                              | Difference<br>in mean<br>(2) | % reduction<br> Bias <br>(3) | Difference<br>in mean<br>(4) | % reduction<br> Bias <br>(5) | Difference<br>in mean<br>(6) | % reduction<br> Bias <br>(7) | Difference<br>in mean<br>(8) | % reduction<br> Bias <br>(9) |
| Order Pay | 397.69***                    | -42.24                       | 89.7                         | -16.86                       | 95.8                         | -6.87                        | 98.3                         | 8.18                         | 97.9                         |
| Cainiao   | 0.0016                       | -0.0044                      | -162.6                       | -0.0040                      | -155.7                       | -0.0035                      | -124.1                       | -0.0061                      | -141.1                       |
| ATE       |                              | -0.164***                    |                              | -0.135***                    |                              | -0.128***                    |                              | -0.123***                    |                              |
| ATT       |                              | -0.123***                    |                              | -0.120***                    |                              | -0.112***                    |                              | -0.114***                    |                              |
| ATU       |                              | -0.166***                    |                              | -0.136***                    |                              | -0.128***                    |                              | -0.123***                    |                              |

Note : Matching performed at caliper 0.1

\*p<0.05; \*\*p<0.01; \*\*\*p<0.001

## A.2. Simultaneity

The endogeneity issue due to simultaneity occurs when a change in the output variable causes one of the covariates to change. The simultaneity can occur in our context when sellers react to the historical ratings provided by the customer and improve the delivery time for their future orders. So, we examine the question: Do sellers (and carriers) improve the delivery performance of a given customer order based on past feedback from the same customer?. If this direction of relationship does not exist, we can rule out the possibility of

endogeneity due to simultaneity. We run the following linear model on the data filtered for customers who have a history of multiple orders for a seller and carrier.

$$\text{Log}(\text{Delivery time}_{cijt}) = \alpha + \beta \cdot \text{Past Logistics Rating}_{cijt} + \mathbf{X}_{cijt} \cdot \boldsymbol{\gamma} + \epsilon_{cijt}$$

$\text{Delivery time}_{cijt}$  is the delivery time of customer “ $c$ ” order placed at time “ $t$ ” from seller “ $i$ ” delivered by carrier “ $j$ ”.  $X_{cijt}$  includes all the covariates of the current customer order, promise speed status, customer, seller, carrier fixed effects, and all time fixed effects.  $\text{Past Logistics Rating}_{cijt}$  is the past (or mean) of prior logistics ratings of the same customer  $c$ ’s orders placed from seller “ $i$ ”, carrier “ $j$ ” prior to “ $t$ ”. Table 16 lists the linear regression results. The “no-promise speed” status is set as the reference level for both the models. From columns (1) and (2), we find the coefficients of Logistics rating and Average logistics rating (both being variants of *Past Logistics Rating*) are positive and insignificant. This indicates that although Cainiao or sellers make efforts to shorten the delivery time for customers whose logistics ratings from their previous orders drop, it is not statistically significant. As a result, we can rule out the possibility of endogeneity due to simultaneity. To conclude, we note that if the endogeneity in our data generating process is due to simultaneity, the model specification (Heckman ordered probit model) in the main paper provides causal effects of delivery time on logistics rating.

**Table 16 Linear Regression results**

| Variable |                                          | Log(Delivery time)   |                      |
|----------|------------------------------------------|----------------------|----------------------|
|          |                                          | (1)                  | (2)                  |
|          | Logistics rating (lag 1)                 | 0.017<br>(0.011)     |                      |
|          | Average logistics rating (lag 1 & lag 2) |                      | 0.018<br>(0.015)     |
|          | Pay norm                                 | 0.000*<br>(0.000)    | 0.000<br>(0.000)     |
|          | Two-day promise speed                    | −0.338***<br>(0.004) | −0.328***<br>(0.015) |
|          | One-day promise speed                    | −0.917***<br>(0.008) | −0.854***<br>(0.033) |
|          | Cainiao                                  | −0.024***<br>(0.004) | −0.047***<br>(0.015) |
|          | Controls                                 |                      |                      |
|          | Customer, Seller, Carrier                | Yes                  | Yes                  |
|          | Week, Day, Hour                          | Yes                  | Yes                  |
|          | Holidays                                 | Yes                  | Yes                  |
|          | Observations                             | 182,885              | 14,829               |
|          | Adjusted R <sup>2</sup>                  | 41.9%                | 39.4%                |
|          | F Statistic                              | 336.284***           | 26.289***            |

\*p<0.05; \*\*p<0.01; \*\*\*p<0.001

Clustered standard errors at seller \* carrier level

### A.3. Omitted Variable Bias

The endogeneity due to omitted variable bias occurs when an unobserved variable affects both the output variable and one of the covariates in the model specification. The first step towards addressing the omitted variable or endogeneity problem of any kind is to justify its theoretical existence (Ketokivi and McIntosh 2017). The key covariate of interest in our main econometric specification (Heckman ordered probit model) is delivery time. If there are any unobserved variables in the Heckman ordered probit model that affects delivery time and logistics rating, the coefficient estimates of the delivery time can be biased and inconsistent. For example, failing to include the seller (or carrier) fixed effect can potentially lead to bias in the coefficients. Sellers are heterogeneous in their shipment policies of products from their warehouse or intermediate stations, which affects the package's delivery time. As a result, the delivery time is correlated with the seller.

Further, customers can be heterogeneous in quality perceptions of sellers (Li and Xiao 2010). Cui et al. (2019) shows that removing a high-quality logistics carrier option for a large online retailer leads to a decrease in sales by 16.42%, while its resumption results increase in sales by 18.83%. As a result, a customer's likelihood to post a rating and the value of rating they provide can be correlated with the seller. The Heckman ordered probit model in the revised version includes all exhaustive controls. The time-invariant controls comprise seller and carrier fixed effects. The time effects comprise week of the month, day of week, hour of the day, and holidays. Additional controls include a variant of order amount paid and channel (Cainiao or seller). Our model specifications contain exhaustive controls within the finest granularity of the data, thus limiting any endogeneity problems caused due to omitted variables.

**Instrumental Variable Analysis.** Instrumental variable methods are widely used to estimate causal effects in non-experimental observational data (Wooldridge 2010). The primary factor for these methods' success is to find a valid instrumental variable(s) that satisfy both relevance and exclusion restriction conditions (Wooldridge 2010). For relevance, the instrument should be correlated with the endogenous variable (delivery time), and for exclusion, it should be uncorrelated with the dependent variable (logistics rating). We identified two instruments: (i) seller utilization, (ii) carrier utilization, both varying by the hour of the day in our data. From a queuing perspective, delivery time is equivalent to the total time customer waits in the service system from the moment they placed the order to receipt of the package. As a result, seller and carrier utilization are likely to impact delivery time, satisfying the relevance condition (Wooldridge 2010, Kulkarni 2016). The seller or carrier utilization is unlikely to impact the logistics ratings as these measures are not directly observed by the customers satisfying the exclusion restriction condition. We employed a two-stage residual inclusion approach (Terza et al. 2008, Wooldridge 2015) using these two instruments. In the first stage, we ran a linear model with the logarithm of delivery time as a function of seller utilization, carrier utilization, and remaining exogenous variables - a variant of order amount paid, seller, carrier fixed effects, channel, and time fixed effects. As expected, we found both the coefficients of utilization measures positive and statistically significant ( $p < 0.001$ ), indicating that delivery time increases with utilization. However, the incremental  $R^2$  due to these instruments was less than 1% as compared to an  $R^2$  of around 40% for the complete model, leaving us with extremely weak instruments. Prior research notes that weak instruments can cause more bias in the two-stage model estimates than OLS estimates (Larcker and Rusticus 2010, Rossi 2014). As a result, the instrumental variable method is not feasible in our context to establish causal effects.

In summary, we conclude that if the endogeneity in our data generating process is due to selection bias (selection on observables and unobservables), we establish causal effects of delivery performance on logistics ratings by employing quasi-experimental methods such as Propensity Score Matching and Regression Discontinuity Design. On the contrary, if the endogeneity is due to simultaneity and omitted variable bias, we rule out the possibility or minimize the extent to which these problems can cause bias in our estimates.

## **B. Data Robustness**

We utilized customer order and logistics data from the Tmall platform and Cainiao network observed during April 2017 to examine the research questions stated in the main paper. We employed the Heckman ordered probit regression model on this data to examine the first phase (Q1a) of the main research question. We investigate the robustness of these results (estimates direction and significance) by employing the same model on customer order data observed in a different month, i.e., May 2017. Table 17 lists the regression results for this data set. We find that the coefficient estimates are similar in direction and significance as compared to the main results (Table 3), establishing the robustness of our results and insights on a data set that is different than in the main paper.

Table 17 Heckman Ordered Probit Regression results - May 2017

| Variable        | No-promise speed   |                      |          | Two-day promise speed |                      |          | One-day promise speed |                      |          |
|-----------------|--------------------|----------------------|----------|-----------------------|----------------------|----------|-----------------------|----------------------|----------|
|                 | Probit Selection   | Ordered Probit       | Variable | Probit Selection      | Ordered Probit       | Variable | Probit Selection      | Ordered Probit       | Variable |
| (1)             | (2)                | (3)                  | (4)      | (5)                   | (6)                  | (7)      | (8)                   | (9)                  |          |
| 2 days          | 0.027<br>(0.017)   | -0.033<br>(0.040)    | 2 days   | 0.303<br>(0.036)      | 0.000<br>(0.096)     | 2 days   | -0.029<br>(0.032)     | -0.417***<br>(0.038) |          |
| 3 days          | 0.049**<br>(0.017) | -0.177***<br>(0.040) | 3 days   | 0.043<br>(0.036)      | -0.341***<br>(0.094) | ≥ 3 days | -0.015<br>(0.032)     | -0.640***<br>(0.072) |          |
| 4 days          | 0.079*<br>(0.017)  | -0.261**<br>(0.040)  | ≥ 4 days | 0.009<br>(0.037)      | -0.662***<br>(0.098) |          |                       |                      |          |
| 5 days          | 0.101**<br>(0.018) | -0.346***<br>(0.041) |          |                       |                      |          |                       |                      |          |
| ≥ 6 days        | 0.069*<br>(0.019)  | -0.513***<br>(0.042) |          |                       |                      |          |                       |                      |          |
| Pay norm        | 0.007*<br>(0.003)  | 0.033***<br>(0.006)  |          | 0.006*<br>(0.003)     | 0.028***<br>(0.005)  |          | -0.006<br>(0.004)     | 0.011<br>(0.013)     |          |
| Cainiao         | 0.014<br>(0.021)   | -0.010<br>(0.015)    |          | -0.000<br>(0.013)     | -0.035<br>(0.014)    |          | -0.002<br>(0.021)     | 0.006<br>(0.030)     |          |
| $\rho$          | 0.000<br>(0.055)   |                      |          | -0.005<br>(0.212)     |                      |          | -0.022<br>(0.479)     |                      |          |
| Seller, Carrier | Yes                | Yes                  |          | Yes                   | Yes                  |          | Yes                   | Yes                  |          |
| Week, Day, Hour | Yes                | Yes                  |          | Yes                   | Yes                  |          | Yes                   | Yes                  |          |
| Holidays        | Yes                | Yes                  |          | Yes                   | Yes                  |          | Yes                   | Yes                  |          |
| N               | 14,942,463         | 5,551,858            |          | 1,378,656             | 509,976              |          | 142,333               | 52,383               |          |
| LL              | -12,365,110.7      |                      |          | -1,068,233.0          |                      |          | -107,766.6            |                      |          |

\*p&lt;0.05; \*\*p&lt;0.01; \*\*\*p&lt;0.001

Clustered standard errors at seller \* carrier level